

# $\Pi$ : A Unified Framework for Computational Verifiable Secret Sharing

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**Abstract.** An  $(n, t)$ -Verifiable Secret Sharing (VSS) scheme allows a dealer to share a secret among  $n$  parties, s.t. all the parties can verify the validity of their shares and only a set of them, i.e., more than  $t$ , can access the secret. In this paper, we present  $\Pi$ , as a unified framework for constructing computational VSS schemes in the honest-majority setting, based on Shamir secret sharing. Notably,  $\Pi$  does not rely on homomorphic commitments; instead requires a random oracle and any commitment scheme that extra to its core attributes hiding and binding, it might be homomorphic and/or Post-Quantum (PQ) secure.

- (i) When employing Discrete Logarithm (DL)-based commitments,  $\Pi$  enables the construction of two novel VSS schemes in the RO model, named  $\Pi_P$  and  $\Pi_F$ . Compared to the well-known Pedersen and Feldman VSS schemes, both  $\Pi_P$  and  $\Pi_F$  require  $O(1)$  (resp.  $O(t)$ ) exponentiations in the verification (resp. reconstruction) process, as opposed to  $O(t)$  (resp.  $O(t^2)$ ), albeit at the expense of a constant factor slower sharing and increased communication.
- (ii) By instantiating  $\Pi$  with a hash-based commitment, we obtain a novel PQ-secure VSS scheme, labeled  $\Pi_{LA}$  (pronounced [paɪˈla]<sup>1</sup>).  $\Pi_{LA}$  outperforms the recent protocol by Atapoor, Baghery, Cozzo, and Pedersen from Asiacrypt’23 by a constant factor in *all* metrics.  $\Pi_{LA}$  can also be seen as an amplified version of the *simple* VSS scheme, proposed by Gennaro, Rabin, and Rabin at PODC’98.
- (iii) Building upon  $\Pi_F$ , we construct a Publicly VSS (PVSS) scheme, labeled  $\Pi_S$ , that can be seen as a new variant of Schoenmakers’ scheme from Crypto’99. To this end, we first define the Polynomial Discrete Logarithm (PDL) problem, as a generalization of DL and then build a variant of the Schnorr Proof of Knowledge (PoK) scheme based on the new hardness assumption. We think the PDL relation and the associated PoK scheme can be independently interesting for Shamir-based threshold protocols.

We believe  $\Pi$  is general enough to be employed in various contexts such as lattices, isogenies, and an extensive array of practical use cases.

**Keywords:** Verifiable Secret Sharing · Polynomial Discrete Logarithm

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<sup>1</sup> In Turkish, ‘Pay’ (pronounced [paɪ]) is a noun for *Share* and ‘Payla’ means *Share it*.

## 1 Introduction

Secret sharing schemes have become foundational tools in threshold cryptography and secure multi-party computation. These schemes facilitate the secure distribution of sensitive information among multiple parties, allowing only qualified shareholders to reconstruct the original secret collaboratively.

Traditional secret sharing schemes, like Shamir’s protocol [26], assume the presence of honest parties but lack provisions for security against malicious ones. To address this concern, Verifiable Secret Sharing (VSS) schemes [14, 16] have been developed, aiming to withstand various attacks, including incorrect share distribution by the dealer and malicious behavior by parties during the reconstruction phase. A Non-Interactive VSS (NI-VSS) scheme allows a dealer to non-interactively (in the happy path) distribute a secret among  $n$  parties, such that all the parties can verify the validity of their shares, and similar to a typical secret sharing scheme, only a specific number of them can access the secret. Numerous VSS schemes are built on regular secret-sharing protocols, adding verifiability features on top [1, 4, 14, 16, 19–21, 23, 25]. Many of known VSS schemes like Feldman [16] and Pedersen [23] use Shamir secret sharing and exploit the homomorphic property of the Discrete Logarithm (DL) and Pedersen commitment to achieve verifiability. To this end, the dealer sends the shares securely to parties and publishes the homomorphic commitments to the coefficients of the underlying secret polynomial. Then, they leverage the homomorphic property of the DL group to convince the shareholders that the secret sharing is performed correctly. Publicly Verifiable Secret Sharing (PVSS) schemes additionally allow an external verifier to verify the validity of the distributed shares (that are encrypted under the public key of the shareholders) in a single round [9, 18, 21, 25, 28].

In [19, Section 2], Gennaro, Rabin, and Rabin (GRR) proposed a *simple* VSS scheme for  $n \geq 2t + 1$  that does not need homomorphic commitments. However, their construction achieves a weaker security in terms of reconstruction. Namely, from the  $n$  distributed shares, any different  $t + 1$  honest shareholders might reconstruct a different secret. The reason is that in their construction [19, Fig. 1], the dealer does not prove that all the shares are generated using a unique degree- $t$  polynomial  $f(X)$ . To deal with this concern, they propose an amplified version of their simple construction, that uses homomorphic commitments (i.e., Pedersen commitment) and achieves the stronger notion of extractability, which guarantees that any different  $t + 1$  honest shareholders reconstruct a unique secret  $f(0)$ . The VSS schemes in plain model, which are based on homomorphic commitments, e.g., [4, 16, 19, 23], have at the best  $O(t\lambda)$  communication complexity and require  $O(n)$  or  $O(t)$  exponentiations, in the sharing and verification sides, respectively, where  $\lambda$  denotes the security parameter. In [2, Section 3.1], Backes, Kate, and Patra proposed the first VSS scheme for  $n \geq 2t + 1$ , that do not require homomorphic commitments. However, their construction uses bivariate polynomials [4] to achieve verifiability, that requires  $O(n^2)$  commitments and  $O(n^2\lambda)$  bits of broadcast,  $O(n^2\lambda)$  bits of private communication in the sharing phase and also imposes  $O(n^2\lambda)$  broadcasts in the reconstruction phase. In an elegant recent work, Atapoor, Bagheri, Cozzo, and Pedersen (ABCP) [1] intro-

duced the first Post-Quantum (PQ) secure VSS scheme for  $n \geq 2t + 1$  which uses a quantum Random Oracle (RO) and a (collapsing) hash-based commitment scheme and boasts computational and communication costs of  $O(n)$  and  $O(n\lambda)$ , respectively. Notably, their scheme relies solely on *lightweight* operations, such as hashing and polynomial evaluations, making it significantly more efficient than previous schemes in this setting. In a different setting, Shoup and Smart [27] also recently unveiled a novel lightweight asynchronous VSS scheme that similarly employs a random oracle (or a random beacon) and *lightweight* cryptographic operations, specifically hashing and polynomial evaluations. Shoup and Smart’s scheme is tailored for the asynchronous communication model and necessitates at least  $2/3$  of the participants to be honest. Both the mention works [1, 27] have used hash-based commitments to build PQ secure and lightweight VSS schemes. Our study is in the synchronous setting, assumes that the majority of parties are honest, and aims to harness the strengths of both lightweight and heavyweight cryptography. More precisely, we aim to construct new VSS schemes that either improve existing constructions in terms of efficiency or by sacrificing PQ security (and lightweighness), can achieve unique features such as Information-Theoretical (IT) unpredictability (such as Pedersen VSS [23]) or public verifiability (as in Schoenmakers PVSS [25]). These features cannot be achieved in VSS schemes that use hash-based commitments.

The starting point for ABCP [1] is the construction of a PQ-secure Non-Interactive Threshold Zero-Knowledge (NI-TZK) proof scheme for the following  $n$ -distributed relations  $R_1, \dots, R_n$ :

$$R_i = \{(f_i, f(X)) | f(i) = f_i\}, \quad i = 1, \dots, n. \quad (1)$$

Here  $f(X)$  represents a witness polynomial in  $X$  of degree (at most)  $t$  with coefficients defined over the ring  $\mathbb{Z}_N$ , and  $f_i$  are the shares received by  $n$  parties.

NI-TZK proofs are formally defined and studied by Boneh et al. [7]. In a NI-TZK proof scheme, a prover aims to convince  $n$  verifiers, holding a piece of the statement, e.g.,  $f_i$ , that the main statement, e.g.,  $f_1 \parallel \dots \parallel f_n$  (hidden from an individual verifier) belongs to a specific language. Similar to the typical cases, such proof systems must be complete, meaning that if the main statement is in language, an honest prover will be able to convince honest verifiers. They should satisfy soundness, meaning that if the main statement is not in the language, then all verifiers will reject the verification except for a negligible probability. However, in some cases a subset of verifiers, e.g., up to  $t$  of them, may be malicious and collude with an adversarial prover. Finally, they need to satisfy a variant of ZK, so-called *Threshold ZK* (TZK)<sup>2</sup>, as introduced by Boneh et al. [7]. TZK implies that any subset of the verifiers up to  $t$ , should learn no additional information about the main statement, beyond their own shares of statement and the fact that the main statement belongs to the language.

ABCP [1] coined the term “Shamir relation” to describe the  $n$ -distributed relation in Eq. (1). Then, they used the proposed PQ-secure NI-TZK for the

<sup>2</sup> We adopt the term “Threshold ZK” from [1] to refer this variant of zero-knowledge, and it is called “Strong ZK” in [7].

Shamir relation and built an extremely efficient computationally secure VSS scheme in the honest-majority setting, which uses hash functions and polynomial evaluations. Drawing upon the NI-TZK proofs, they also introduced a new approach for secret reconstruction in VSS schemes. In certain scenarios, this approach can lead to the development of more efficient threshold protocols, such as Distributed Key Generation (DKG) protocols and threshold signatures.

## 1.1 Our Contributions

**$\Pi$ : A Unified Framework for Computational VSS Schemes.** We present a unified framework  $\Pi$  designed for constructing computational VSS schemes in the honest-majority setting. The framework is based on Shamir secret sharing and draws inspiration from the VSS scheme recently introduced by Atapoor, Bagheri, Cozzo, and Pedersen [1] and the *simple* construction presented by Gennaro, Rabin, and Rabin [19]. In its general form,  $\Pi$  uses a (classic or quantum) random oracle, and does not necessarily need a homomorphic commitment scheme (as in Feldman [16] and Pedersen [23] schemes). Nevertheless, the option remains to instantiate it with homomorphic commitments, to construct new VSS schemes that are more efficient than current schemes, or achieve unique properties such as Public Verifiability (PV) [25] and IT unpredictability [23]. At its core, the framework boasts a general and efficient construction, enabling the construction of VSS schemes with diverse features.

*Sharing.* In the main construction of  $\Pi$ , given a secret  $f_0$ , a hiding and binding commitment scheme  $\mathcal{C}$ , and a (classic or quantum) random oracle  $\mathcal{H}$ , the dealer proceeds as follows: 1) Does Shamir secret sharing: samples a random degree- $t$  polynomial  $f(X)$  with free term  $f_0$  and sets  $f_i = f(i)$  for  $i = 1, \dots, n$ . 2) Samples another random degree- $t$  polynomial  $r(X)$  and sets  $r_i = r(i)$  for  $i = 1, \dots, n$ . 3) Sets  $c_i = \mathcal{C}(f_i, r_i)$  (or  $c_i = \mathcal{C}((f_i, r_i), \gamma_i)$ , where  $\gamma_i$  are randomnesses used in the commitment phase) for  $i = 1, \dots, n$  and  $z(X) = r(X) + d \cdot f(X)$ ; where  $d = \mathcal{H}(c_1, \dots, c_n)$ . Finally, securely transmit  $f_i$  (and the randomizer  $\gamma_i$  employed in the commitment  $\mathcal{C}$ , if applicable) to party  $P_i$ , and publish  $\pi_{share} := (z(X), c_1, \dots, c_n)$ . In general, sharing can be as efficient as performing two Shamir secret sharing in addition to  $n$  commitments.

*Verification.* In the general form, given  $(f_i, z(X), \{c_i\}_{i=1}^n)$ , to verify the received share  $f_i$  (and  $\gamma_i$  if applicable), party  $P_i$  checks if  $z(X)$  is a degree- $t$  polynomial, computes  $d = \mathcal{H}(c_1, \dots, c_n)$  and checks if  $c_i = \mathcal{C}(f_i, z(i) - d \cdot f_i)$  (or  $c_i = \mathcal{C}((f_i, z(i) - d \cdot f_i), \gamma_i)$ ). If the checks do not pass,  $P_i$  broadcasts a complaint against the dealer. If player  $P_j$  broadcasted a complaint, then the dealer broadcasts the share  $f_j$  (and  $\gamma_j$  if applicable), such that  $c_j = \mathcal{C}(f_j, z(j) - d \cdot f_j)$  (or  $c_j = \mathcal{C}((f_j, z(j) - d \cdot f_j), \gamma_j)$ ). If the dealer does not follow the protocol, he is disqualified, otherwise the protocol continues as usual.

*Reconstruction.* Each shareholder broadcasts their shares  $f_i$  (and  $\gamma_i$  if applicable). The disclosed shares are verified using the verification process of the target VSS scheme. Following verification,  $t + 1$  valid shares are used for the reconstruction of the secret polynomial  $f(X)$ , resulting in the secret  $s = f(0)$ .

*Security.* We show that, given a secure commitment scheme, in the honest-majority scenario where  $t + 1$  of the parties are honest, with  $n \geq 2t + 1$ , the general construction satisfies *verifiability* (implied by soundness against prover and  $t$  malicious verifiers), and unpredictability or simulatability (implied by threshold zero-knowledge) in the random oracle model. These properties ensure that a malicious dealer cannot convince honest parties except with a negligible probability. Consequently, any set of  $t + 1$  honest parties will be able to collectively reconstruct a *unique* secret  $f_0 = f(0)$  in the reconstruction phase.

It's crucial to mention that when we use  $\Pi$  to build VSS schemes aiming to satisfy unpredictability, the dealer commits to *random* values  $\{f_i, r_i\}_{i=1}^n$ , where possess sufficient entropy. Consequently, there is no need for an additional randomizer in the commitment process, even if  $z(i) = r_i + d \cdot f_i$  and  $d$  are public. However, when we use  $\Pi$  to build an VSS scheme that satisfies IT simulatability, like Pedersen scheme, i.e., in case the secret  $f_0$  does not have enough entropy, the dealer must use an additional randomizer in the commitment. Then, the use of a separate randomizer, i.e.,  $\gamma_i$ , becomes essential. As can be seen, the strength of the new framework lies in its simplicity and generality and it is flexible enough to be tailored for constructing various VSS schemes with distinct properties. In addition to Shamir's secret sharing, it only requires a secure commitment scheme and a classic/quantum RO. Commitment schemes and classic/quantum RO can be efficiently constructed respectively using various fundamental primitives (such one-way functions) and hash functions, rendering  $\Pi$  a unified framework for building VSS schemes that can also achieve IT simulatability, public verifiability and/or PQ security. Leveraging  $\Pi$ , we present a range of novel and efficient VSS schemes that, in general, can outperform current alternatives, particularly in the verification and reconstruction phases.

**New VSS Schemes from DL-Based (Homomorphic) Commitments.** By instantiating  $\Pi$  with standard Pedersen commitment scheme [23], we introduce an efficient alternative for the well-known Feldman scheme [16], labeled as  $\Pi_F$ . When dealing with an extension of Pedersen commitment (i.e., using three random generators, instead of two)  $\Pi$  enables the construction of a novel IT-secure VSS scheme referred to as  $\Pi_P$ . Similar to the Pedersen scheme, in  $\Pi_P$ , fewer than  $t$  parties learn nothing about the main secret, ensuring IT simulatability.

In terms of efficiency, both  $\Pi_F$  and  $\Pi_P$  require  $O(1)$  (resp.  $O(t)$ ) exponentiations in the verification (resp. reconstruction) process, as opposed to  $O(t)$  (resp.  $O(t^2)$ ) in the Feldman [16] and Pedersen [23] schemes, where  $t$  represents the threshold parameter. This improvement comes at the cost of a constant factor of overhead in the sharing phase and communication and using a random oracle.

**More Efficient VSS Scheme from Hash Functions.** Through the instantiation of  $\Pi$  using a quantum RO and a non-homomorphic commitment scheme, such as those based on hash functions, we obtain a novel PQ-secure VSS scheme, named  $\Pi_{LA}$ . It can be viewed as an alternative to the recent scheme by ABCP [1], which similarly employs a quantum RO and a hash-based commitment scheme. Compared to the ABCP VSS scheme [1],  $\Pi_{LA}$  outperforms by constant factor in terms of all efficiency metrics. From a different view,

$\Pi_{\mathbf{LA}}$  can be seen as an amplified version of the *weak* VSS scheme proposed by GRR in [19, Section 2], as their *simple* construction also uses non-homomorphic commitments to commit  $(f_i, r_i)$  for  $i = 1, \dots, n$ . However, compared to their *weak* VSS scheme,  $\Pi_{\mathbf{LA}}$  satisfies the stronger notion of constructability, i.e., any different set of  $t + 1$  honest parties will reconstruct a *unique* secret.

### Generalizing DL Relation and Schnorr’s Protocol Over Polynomials.

The well-known Schnorr ID protocol [24] allows one to prove knowledge of witness for the relation  $R_{DL} = \{(g, F), f \mid F = g^f\}$  where  $g$  is the group generator,  $f \in \mathbb{Z}_q$  is the witness value, which can also be interpreted as a degree-0 polynomial with a single coefficient defined over  $\mathbb{Z}_q$ . In Sec. 5, we generalize  $R_{DL}$  relation over polynomials and introduce the Polynomial Discrete Logarithm (PDL) relation denoted as  $R_{PDL}$ , which is defined as follows,

$$R_{PDL} = \{(g, x_i, F_i), f(X) \mid F_i = g^{f(x_i)}\} \text{ for } i = 1, 2, \dots, n.$$

Here,  $f(X) \in \mathbb{Z}_q[X]_t$  is a (at most) degree  $t \leq n-1$  witness polynomial with coefficients from  $\mathbb{Z}_q$ , and  $\{x_i\}_{i=1}^n$  are  $n$  *distinct* elements from  $\mathbb{Z}_q$ . Then, we present a Non-Interactive Zero-Knowledge (NIZK) Proof-of-Knowledge (PoK) scheme  $\pi_{PDL}$  based on Schnorr’s protocol, that allows a prover to prove knowledge of a witness for  $R_{PDL}$  relation. We believe  $\pi_{PDL}$  can be a useful proof scheme for constructing threshold protocols based on Shamir secret sharing, specifically for  $n \geq 2t + 1$  and  $x_1 = 1, \dots, x_n = n$ .

To the best of our knowledge, this marks the first explicit definition of the PDL problem, even though it has been implicitly employed in certain prior VSS schemes and protocols [9, 10, 16, 25]. In [10], Cascudo and David similarly defined a slightly different variant of  $R_{PDL}$  and built a sigma protocol for this variant. However, there are some issues in their proposed sigma protocol, which will be discussed in detail in Sec. 5. They also presented a probabilistic verification protocol for the  $R_{PDL}$  relation, which achieves soundness, in contrast to our proposed NIZK proof of knowledge  $\pi_{PDL}$ . A detailed comparison of our construction with theirs is provided later in this paper.

**A Novel PVSS Scheme Based on DL.** Using the new NIZK PoK scheme  $\pi_{PDL}$ , and building upon a variant of VSS scheme  $\Pi_{\mathbf{F}}$ , we introduce a novel Publicly Verifiable Secret Sharing (PVSS) scheme, designated as  $\Pi_{\mathbf{S}}$ .  $\Pi_{\mathbf{S}}$  serves as a more efficient alternative to Schoenmakers’ PVSS scheme from Crypto’99 [25]. Compared to Schoenmakers’ scheme [25],  $\Pi_{\mathbf{S}}$  streamlines the verification complexity from  $O(nt)$  to  $O(n)$ , accelerates the sharing phase by more than two times, and reduces the communication cost slightly. In essence,  $\Pi_{\mathbf{S}}$  improves all efficiency metrics in Schoenmakers’ scheme with no additional expense. In [9, 10], Cascudo and David have proposed different variants of Schoenmakers’ PVSS scheme, all reducing the verification complexity to  $O(n)$ . In comparison to their schemes from [9],  $\Pi_{\mathbf{S}}$  generally demonstrates superior efficiency. Notably, its verification process is at least  $3 - 4\times$  faster than the verification of their schemes. In their later work [10], Cascudo and David extended and optimized their RO-based scheme from [9] to support packed secret sharing. Notably, we found that

$\Pi_S$  shares similarities with the unpacked case of their scheme [10]. Using optimization employed in  $\Pi_S$ , the unpacked version of their scheme can achieve the same performance to  $\Pi_S$ . It is important to note that  $\Pi_S$  is developed in a generic manner, with its security reduced to the PDL problem, providing a clearer and simpler security proof. The scheme by Cascudo and David [10] relies on a sigma protocol tailored for a variant of the  $R_{PDL}$  relation. However, in their security proof of sigma protocol [10, Proposition 1], there is a lack of a clear reduction to a hardness assumption, and their security proof for special soundness lacks an extraction algorithm. We elaborate more on this matter later in Sec. 5. We highlight that PVSS schemes can generally be categorized into two families, each with different applications: The VSS schemes where the secret is  $g^{f_0}$  and the schemes where the secret is  $f_0$  [11]. The PVSS scheme  $\Pi_S$  belongs to the first category, where each party obtains a share  $g^{f_i}$  and latter during the reconstruction phase, the parties collectively reconstruct  $g^{f_0}$ .

*Efficiency Comparisons of New Schemes.* Table 1 provides a summary of performance metrics for the proposed VSS and PVSS schemes, including  $\Pi_F$ ,  $\Pi_P$ ,

**Table 1.** A comparison of new VSS and PVSS schemes with those of Feldman [16], Pedersen [23], Schoenmakers [25], Cascudo-David [9,10], and ABCP [1]. Commu.: Communication, DL: Discrete Logarithm, PDL: Polynomial Discrete Logarithm, DDH: Decisional Diffie-Hellman, DBS: Decisional Bilinear Square, IT-S: Information Theoretically Simulatability, Classic: Classical security, PQ: Post-quantum security, RO: Random Oracle, Plain: Plain Model, BC: Broadcast,  $n$ : Number of parties,  $t$ : threshold parameter ( $t \approx n/2$ ),  $P_G$ : Pairing Operation,  $E_G$ : Exponentiation in group  $G$ ,  $M_G$ : Multiplication in group  $G$ ,  $\mathcal{PE}$ : degree- $t$  Polynomial Evaluation,  $\mathcal{H}$ : Hashing,  $|G|$ :  $G$  element size,  $|Z_q|$ :  $Z_q$  element size,  $|Z_N|$ :  $Z_N$  element size,  $|\mathcal{H}|$ : Output size of  $\mathcal{H}$ .

| (P)VSS & Security                                 | Sharing                               | Dealer's Commu.                                         | Verification                                  | Reconstruct                                    |
|---------------------------------------------------|---------------------------------------|---------------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| Feldman [16]<br>(DL, Plain, Classic)              | $0.5n E_G$<br>$1n \mathcal{PE}$       | Private: $1n Z_q $<br>BC: $0.5n G $                     | $t E_G +$<br>$t M_G$                          | $t^2 E_G +$<br>$t^2 M_G$                       |
| Sec. 4.1, $\Pi_F$<br>(DL, RO, Classic)            | $2n E_G$<br>$2n \mathcal{PE}$         | Private: $1n Z_q $<br>BC: $n G  + 0.5n Z_q $            | $2 E_G +$<br>$1 \mathcal{PE} + 1 \mathcal{H}$ | $2t E_G +$<br>$t \mathcal{PE} + t \mathcal{H}$ |
| Pedersen [23]<br>(DL, Plain, IT-S)                | $1n E_G$<br>$2n \mathcal{PE}$         | Private: $2n Z_q $<br>BC: $0.5n G $                     | $t E_G$<br>$+ t M_G$                          | $t^2 E_G$<br>$+ t^2 M_G$                       |
| Sec. 4.2, $\Pi_P$<br>(DL, RO, IT-U)               | $3n E_G$<br>$2n \mathcal{PE}$         | Private: $2n Z_q $<br>BC: $n G  + 0.5n Z_q $            | $3 E_G +$<br>$1 \mathcal{PE} + 1 \mathcal{H}$ | $3t E_G +$<br>$t \mathcal{PE} + t \mathcal{H}$ |
| ABCP [1]<br>(Hash, RO, PQ)                        | $2n \mathcal{H}$<br>$2n \mathcal{PE}$ | Private: $1n Z_N $<br>BC: $2n \mathcal{H}  + 0.5n Z_N $ | $1 \mathcal{PE}$<br>$+ 3 \mathcal{H}$         | $t \mathcal{PE}$<br>$+ 3t \mathcal{H}$         |
| Sec. 4.3, $\Pi_{LA}$<br>(Hash, RO, PQ)            | $1n \mathcal{H}$<br>$2n \mathcal{PE}$ | Private: $1n Z_N $<br>BC: $n \mathcal{H}  + 0.5n Z_N $  | $1 \mathcal{PE} +$<br>$+ 2 \mathcal{H}$       | $t \mathcal{PE} +$<br>$+ 2t \mathcal{H}$       |
| Sch. [25]-PVSS<br>(DDH, RO, Classic)              | $4.5n E_G$<br>$1n \mathcal{PE}$       | Private: —<br>BC: $1.5n G  + n Z_q $                    | $nt + 4n E_G$<br>$+ 2.5n M_G$                 | $5t E_G$<br>$+ t M_G$                          |
| Cas-Dav [9]-PVSS<br>(DBS, Plain, Classic)         | $2n E_G$<br>$1n \mathcal{PE}$         | Private: —<br>BC: $2n G $                               | $2n P_G +$<br>$n E_G + n M_G$                 | $2t P_G +$<br>$t E_G + t M_G$                  |
| Cas-Dav [9]-PVSS<br>(DDH, RO, Classic)            | $4n E_G$<br>$1n \mathcal{PE}$         | Private: —<br>BC: $2n G  + n Z_q $                      | $5n E_G$<br>$+ 3n M_G$                        | $5t E_G$<br>$+ t M_G$                          |
| Sec. 6, $\Pi_S$ & [10]<br>(PDL, DDH, RO, Classic) | $2n E_G$<br>$2n \mathcal{PE}$         | Private: —<br>BC: $n G  + 0.5n Z_q $                    | $2n E_G +$<br>$n \mathcal{PE} + n M_G$        | $5t E_G +$<br>$t M_G$                          |



and  $\Pi_{\text{LA}}$ , as well as the PVSS scheme  $\Pi_{\text{S}}$ . These metrics are compared with relevant schemes from the literature [1, 9, 10, 16, 23, 25]. In [22], authors proposed a VSS scheme in the Common Reference String (CRS) and RO models, which can have  $O(\lambda)$  online communication size. However, their scheme relies on strong assumptions in bilinear groups and requires a trusted CRS of size  $O(n\lambda)$ .

## 1.2 Implications of New Results

We expect any cryptographic construction that uses either of the VSS schemes of Feldman [16], Pedersen [23], ABCP [1], or PVSS schemes [9, 10, 21, 25], or a variation of them, can be potentially affected by the new results. Considering NIST’s Threshold Cryptography project<sup>3</sup>, which seeks to standardize threshold schemes for cryptographic primitives, we think the implications of our results can extend beyond theoretical implications, offering practical promise for improving real-world (threshold) cryptographic systems. Delving into the details of revisiting concrete threshold protocols lies beyond the scope of this paper. It is worth noting that the practical usage of new VSS schemes (i.e.,  $\Pi_{\text{F}}$ ,  $\Pi_{\text{P}}$ , and  $\Pi_{\text{LA}}$ ) differs significantly from previous schemes. In practice, using the new VSS schemes might involve constructing a NI-TZK proof for specific languages tailored for a target application. An example construction for such NI-TZK proofs can be found in [1, Sec. 4.1].

In this vein, our generalized variant of Schnorr’s NIZK PoK scheme to the PDL relation (defined in Eq. (3)) can be a useful tool for constructing threshold cryptographic protocols based on Shamir secret sharing. Notably, the idea behind it is general enough for versatile deployment across PQ secure contexts.

## 1.3 Outline

In Sec. 2, we present an overview of some preliminary concepts. In Sec. 3, we introduce the new framework  $\Pi$  devised for constructing VSS schemes. Leveraging  $\Pi$ , in Sec. 4 we present several new VSS schemes, with different features. In Sec. 5, we generalize DL problem and Schnorr protocol over polynomials and present an efficient NIZK PoK scheme, that can be a useful tool for  $\Pi$ , while also preserving potential interest for different purposes. In Sec. 6, we present an efficient PVSS scheme. Finally, we conclude the paper in Sec. 7.

# 2 Preliminaries

## 2.1 Notation, Fields, Groups, Exceptional Sets

We let  $\lambda$  denote a security parameter. We use the assignment operator  $\leftarrow$  to denote uniform sampling from a set  $\mathcal{E}$ , e.g.  $x \leftarrow \mathcal{E}$ . Throughout this paper  $p$  and  $q$  denote two large primes such that  $q$  divides  $p-1$ ,  $\mathbb{G}$  is the unique subgroup of  $\mathbb{Z}_p^*$  of order  $q$ , and  $g$  is a generator of cyclic group  $\mathbb{G}$  of prime order  $q$ . One can

<sup>3</sup> More on <https://csrc.nist.gov/Projects/Threshold-Cryptography/>.



test if an element  $a \in \mathbb{Z}_p^*$  is in  $\mathbb{G}$ , by checking if  $a^q = 1$ . The group  $\mathbb{G}$  is chosen such that computing DL of  $h \in \mathbb{G}$ , i.e.,  $\log_g h$ , is hard in this group. We write  $\mathbb{Z}_q$  and  $\mathbb{Z}_q[X]_t$  for polynomials of degree  $t$  in the variable  $X$  and with coefficients in finite field  $\mathbb{Z}_q$ , with known prime  $q$ . When we refer to groups we assume they have known prime order and efficient algorithms to compute group operations. It will be assumed that all parties know  $p$ ,  $q$ ,  $g$ , and  $N$ .

## 2.2 Shamir and Verifiable Secret Sharing

A  $(n, t)$ -Shamir secret sharing scheme [26] allows  $n$  parties to individually hold a share  $f_i$  of a secret  $f_0$ , such that any subset of  $t$  parties or less are unable to learn any information about the secret  $f_0$ , while any subset of at least  $t + 1$  parties are able to efficiently reconstruct the secret  $f_0$ . In more detail, this is achieved via polynomial interpolation over the ring  $\mathbb{Z}_N$  (or field  $\mathbb{Z}_q$ ). A secret polynomial  $f(x) \in \mathbb{Z}_N[x]_t$  is chosen and its free term is set to be the secret  $f_0$ , namely  $f(0) = f_0$ . Each party  $P_i$  for  $i \in \{1, \dots, n\}$  is assigned the secret share  $f_i = f(i)$ . Then any subset  $Q \subseteq \{1, \dots, n\}$  of at least  $t + 1$  parties can reconstruct the secret  $f_0$  via Lagrange interpolation by computing  $f_0 = f(0) = \sum_{i \in Q} f_i \cdot L_{0,i}^Q$ , where  $L_{0,i}^Q := \prod_{j \in Q \setminus \{i\}} \frac{j}{j-i} \pmod{N}$ , are the Lagrange basis polynomials evaluated at 0. Any subset of less than  $t + 1$  parties are unable to find  $f_0 = f(0)$ , as this is information theoretically hidden from the other shares.

If  $\mathbb{Z}_N$  is a ring, the difference of any elements in  $\{1, \dots, n\}$  must be invertible modulo  $N$ , thus  $\{1, \dots, n\}$  must be an exceptional set (defined in Def. 2.1). This is the case if  $n$  is less than the smallest prime divisor  $q$  of  $N$ . In the case where more than  $q$  parties want to participate in the protocol, we would have to work in a subgroup  $\mathbb{Z}_{N'} \subset \mathbb{Z}_N$  such that the smallest divisor of  $N'$  is larger than  $q$ . Next we recall the definition of *(super)exceptional sets*.

**Definition 2.1 (Exceptional set [3, 5, 15]).** An exceptional set (modulo  $N$ ) is a set  $\Xi_k = \{c_1, \dots, c_k\} \subseteq \mathbb{Z}_N$ , where the pairwise difference of all distinct elements is invertible modulo  $N$ . If further the pairwise sum of all elements is invertible modulo  $N$ ,  $\Xi_k$  is called a superexceptional set (modulo  $N$ ).

**Verifiable Secret Sharing.** Standard secret sharing schemes are secure against passive attacks. In many applications, a secret sharing scheme needs to be secure against the malicious dealer or parties with active attacks. This is achieved through VSS schemes, which allow a dealer to share a secret among a group of individuals in a verifiable manner [14]. VSS schemes allow a dealer to distribute the secret in a *verifiable* manner, so that the shareholders can verify the validity of the shares and only a specific number of them can access the secret.

## 2.3 Sigma Protocols

Next, we recall the definition of sigma protocols ( $\Sigma$ -protocols). Here the algorithms are Probabilistic Polynomial-Time (PPT), unless mentioned. Let  $X = X(\lambda)$  and  $W = W(\lambda)$  be sets. Let  $R$  be a relation on  $X \times W$  that defines a

language  $L = \{x \in X : \exists w \in W, R(x, w) = 1\}$ . Given  $x \in L$ , an element  $w \in W$  such that  $R(x, w) = 1$  is called a witness. Let relation generator  $\mathcal{R}$  be a PPT algorithm such that  $\mathcal{R}(1^\lambda)$  outputs pairs  $(x, w)$  such that  $R(x, w) = 1$ .

A public-coin sigma-protocol ( $\Sigma$ -protocol) for the relation  $R$  is a 3-round interactive protocol between two PPT algorithms: a prover  $P$  and a verifier  $V$ .  $P$  holds a witness  $w$  for  $x \in L$  and  $V$  is given  $x$ . In first round,  $P$  sends a commitment value  $a$  to  $V$ , and then in second round,  $V$  answers with a randomly sample challenge value  $d$ . Finally,  $P$  answers with a response  $z$ , and  $V$  verifies the proof and outputs **true** or **false**. The triple  $\text{trans} := (a, d, z)$  is called a transcript of the  $\Sigma$ -protocol. A  $\Sigma$ -protocol is supposed to satisfy *Completeness*, *Honest Verifier Zero-Knowledge* (HVZK), and *Special Soundness* defined below.

**Definition 2.2 (Completeness).** A  $\Sigma$ -protocol with parties  $(P, V)$  is complete for  $\mathcal{R}$ , if for all  $(x, w) \in R$ , the honest  $V$  will always accept the honest  $P$ .

**Definition 2.3 (HVZK).** A  $\Sigma$ -protocol with parties  $(P, V)$  satisfies HVZK for  $\mathcal{R}$ , if there exists a PPT algorithm  $\mathcal{S}$  that given  $x \in X$ , can simulate the  $\text{trans}$  of the scheme, s.t. for all  $x \in L$ ,  $(x, w) \in R$ ,

$$\text{trans}(P(x, w) \leftrightarrow V(x)) \approx \text{trans}(\mathcal{S}(x))$$

where  $\text{trans}(P(\cdot) \leftrightarrow V(\cdot))$  indicates the transcript of the  $\Sigma$ -protocol with  $(P, V)$ , and  $\approx$  denotes the indistinguishability of transcripts.

**Definition 2.4 (Special Soundness).** A  $\Sigma$ -protocol with parties  $(P, V)$  is special sound for  $\mathcal{R}$ , if there exists a PPT extractor  $\mathcal{E}$ , such that for any  $x \in L$ , given two valid transcripts  $(a, d, z)$  and  $(a, d', z')$  for the same message  $a$  but  $d \neq d'$ , then  $\mathcal{E}(a, d, z, d', z')$  outputs a witness  $w$  for the relation  $R$ .

Within the Random Oracle (RO) model, using Fiat-Shamir transform [17], a public-coin, complete, HVZK, and special soundness  $\Sigma$ -protocol can be turned into a Non-Interactive Zero-Knowledge (NIZK) proof or argument of knowledge.

## 2.4 Chaum-Pedersen Protocol for DL Equality

Let  $\mathbb{G}$  be a group with hard DL, and  $g, h$  be two group elements, where  $g$  is the group generator. Let a prover aim to convince a verifier that for the public statement  $g, h, a, b$ , he knows a witness  $x$  which holds in the following relation,

$$R_{DLEQ} = \{(g, h, a, b), x \mid a = g^x \wedge b = h^x\}. \quad (2)$$

This relation is known as DL Equality (DLEQ). In [13], Chaum and Pedersen introduced an efficient NIZK proof of knowledge for DLEQ, as summarized in Fig. 1. This protocol is widely employed in various cryptographic protocols (e.g., threshold decryption, e-voting systems, PVSS schemes, etc.).

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Prover:</b> Given the statement <math>(g, h, a, b) \in \mathbb{G}</math> and the witness value <math>x \in \mathbb{Z}_q</math>, proceed as follows and output a proof <math>\pi</math>.</p> <ol style="list-style-type: none"> <li>1. Sample <math>r \leftarrow \mathbb{Z}_q</math> uniformly at random; and set <math>c_1 = g^r</math> and <math>c_2 = h^r</math>.</li> <li>2. Set <math>d \leftarrow \mathcal{H}(a, b, c_1, c_2)</math>, where <math>\mathcal{H}</math> is a random oracle.</li> <li>3. Set <math>z = r + d \cdot x \pmod q</math>; and Return <math>\pi := (d, z)</math></li> </ol> <p><b>Verifier:</b> Given the statement <math>(g, h, a, b) \in \mathbb{G}</math> and the proof <math>\pi = (d, z)</math>, checks if <math>d = \mathcal{H}(a, b, \frac{g^z}{a^d}, \frac{h^z}{b^d})</math> and outputs <b>true</b> or <b>false</b>.</p> |
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**Fig. 1.** Chaum-Pedersen NIZK proof of knowledge for DLEQ [13].

### 3 A Unified Framework for VSS Schemes

The GRR simple VSS scheme [19] allows a dealer to perform Shamir secret sharing and convince  $n$  verifiers that any  $t + 1$  of them can reconstruct a secret [19]. In their simple scheme, to share  $f_0$ , the dealer first does Shamir secret sharing and obtains the shares  $\{f_i\}_{i=1}^n$ . Then, it samples another degree- $t$  polynomial  $r(X)$  and sets  $r_i = r(i)$  for  $i = 1, \dots, n$ . After that, it commits to  $\{f_i\}_{i=1}^n$  with  $\{r_i\}_{i=1}^n$ , by setting  $c_i = \mathcal{C}(f_i, r_i)$ , where  $\mathcal{C}$  can be any commitment scheme. At the end, it securely sends  $(f_i, r_i)$  to  $P_i$ , and broadcasts  $\{c_i\}_{i=1}^n$ . Although their scheme is highly efficient, it lacks the guarantee of a *unique* reconstructed secret. In certain scenarios, such as robust cloud storage, this lack of uniqueness might not be a concern since computations on the shares are not required. However, when parties aim to perform computations on a unique value  $f(0)$ , they must get sure that they all possess distinct evaluations of a *unique* degree- $t$  polynomial  $f(X)$ . This condition ensures that Lagrange interpolation with any of  $t + 1$  points will lead to a *unique* secret  $f(0)$ . This property, termed *verifiable secret and polynomial sharing*, is described by GRR [19]. To achieve *verifiable secret and polynomial sharing*, the recent PQ-secure VSS scheme by ABCP [1] leverages a hash-based NI-TZK proof scheme for the Shamir relation, which is proven to satisfy *computational* TZK and special soundness in the quantum RO model. In a different setting, the recent lightweight asynchronous VSS scheme by Shoup and Smart [27] also relies on hash-based commitments and either a random beacon or a random oracle to achieve the mentioned property.

In this section, we introduce **II**, designed for constructing VSS schemes with the flexibility to work with both non-homomorphic and homomorphic commitments. It is based on Shamir secret sharing, works in the honest-majority setting and in certain cases (i.e., in case of PVSS scheme) operates on the assumption that each shareholder has registered his/her Public Key (PK), that can facilitate secure communications. **II** combines the strengths of the *simple* VSS scheme from [19], and the standard VSS scheme from [1], to achieve the best of both. Alternatively, it can be viewed as an optimized and generalized version of the ABCP VSS scheme [1]. We note that the authors of [1] specifically targets post-quantum security, which is highlighted in their abstract, introduction, and comparisons. With **II**, we achieve an efficient approach to building VSS schemes with various properties, like perfect (or IT) simulatability and public verifiability.

### 3.1 Our Definitions

Before going through the construction of  $\Pi$ , we summarize our definition of VSS schemes, which are adapted from [1, 23, 25].

**Definition 3.1.** An  $(n, t)$ -VSS consists of four PPT algorithms of (Initial, Share, Verify, Reconstruct) as follows:

1. **Initial:** In this phase, public parameters are sampled and all shared with the parties.
2. **Share** $(n, t, f_0) \rightarrow (\{f_i\}_{i=1}^n, \pi_s)$ : It secret shares  $f_0$  and outputs the shares  $\{f_1, \dots, f_n\}$ , and a (non-interactive) threshold proof  $\pi_{share}$  to prove the validity of the shares. Note that,  $\pi_{share}$  can only be verified by at least  $t + 1$  of the shares (or commitments/encryptions of the shares).
3. **Verify** $(n, t, \{f_i\}_{i=1}^n, \pi_{share}) \rightarrow \text{true/false}$ : Given  $n$ , threshold value  $t$ , the shares  $\{f_i\}_{i=1}^n$  (or commitments/encryptions of them), and the threshold proof  $\pi_{share}$ , generated by **Share**, the algorithm outputs either **true/false**.
4. **Reconstruct** $(\{f_i\}_{i \in Q, |Q|=t+1}, \pi_{share}) \rightarrow \{f_0, \text{false}\}$ : Given  $t + 1$  of the shares and the proof  $\pi_{share}$ , it returns either the secret  $f_0$ , or **false**.

A secure VSS further has two requirements, defined as follows [1, 25].

- **Verifiability constraint:** A shareholder must be able to verify the validity of the received share. If the verification return **true**, then **Reconstruct** should produce a *unique secret*  $f_0$  when run on any  $t + 1$  distinct valid shares. More formally, for any integers  $n \geq 2t + 1$  and  $t \geq 0$ , a VSS is called verifiable if for any PPT adversaries  $\mathcal{A}$ , and for all  $\lambda$ , we have

$$\Pr \left[ (\{f_i\}_{i=1}^n, \pi) \leftarrow \mathcal{A}(n, t), \text{Verify}(n, t, \{f_i\}_{i=1}^n, \pi) = \text{true} : \begin{array}{l} \exists f_0, \forall V \in [n], |V| \geq t + 1, \text{Reconstruct}(\{f_i\}_{i \in V}, \pi) \rightarrow f_0 \end{array} \right] \geq 1 - \text{negl}(\lambda) ,$$

where  $V$  is the set of honest parties.

- **Unpredictability:** The protocol must be unpredictable, meaning that there is no strategy for selecting  $t$  shares of the secret that would enable someone to predict the secret  $f_0$  with a significant advantage. More formally, for any integers  $n > 1$  and  $t < n$ , a VSS is called unpredictable if for all PPT adversaries  $\mathcal{A}$  who has non-adaptively corrupted up to  $t$  parties, for all  $\lambda$ , and a random secret  $f_0$ , we have:

$$\Pr \left[ (\{f_i\}_{i=1}^n, \pi_{share}) \leftarrow \text{Share}(n, t, f_0); Q \subset [n]; |Q| \leq t; \begin{array}{l} f'_0 \leftarrow \mathcal{A}(n, t, \{f_i\}_{i \in Q}, \pi_{share}) : f'_0 = f_0 \end{array} \right] \leq \text{negl}(\lambda) .$$

Our definitions use TZK proofs over shared data [7] to prove the validity of the distributed shares, which their verification requires at least  $t + 1$  honest parties. In some cases, the definition of unpredictability can be strengthened by requiring that the view of adversary who corrupted  $t$  parties can be simulated.

### 3.2 Construction of $\Pi$ and Security Proofs

Let,  $D$  be a dealer and  $P_1, \dots, P_n$  are  $n$  participants of a VSS scheme. Let  $\mathcal{C}$  be a commitment scheme that satisfies both the hiding and binding properties. Specifically, given a commitment  $c = \mathcal{C}(m_b, \gamma)$  and two messages  $(m_0, m_1)$ , it is infeasible or computationally hard to determine which message was committed to (perfect or computational *hiding*). Additionally, it is computationally hard or infeasible to find two distinct pairs  $(m, \gamma)$  and  $(m', \gamma')$  s.t.,  $\mathcal{C}(m, \gamma) = \mathcal{C}(m', \gamma')$  and  $m \neq m'$  (computational or perfect *binding*).

The general construction of  $\Pi$  appears in Fig. 2, which uses a (vector) commitment scheme  $\mathcal{C}$  and a (classic or quantum) random oracle  $\mathcal{H}$ . Intuitively, the general construction employs an efficient and general NI-TZK proof scheme for the Shamir relation, made non-interactive using the Fiat-Shamir transform [17]. It is also important to note that in  $\Pi$ , instead of having separate commitments to the (secret) shares and the randomizers used in the first round of the interactive (threshold) proof schemes, e.g., as in Schnorr protocol (or in ABCP PQ-secure VSS scheme [1]), we use a single (vector) commitment to commit to both (secret) shares (i.e., individual private statements) and the randomizers simultaneously.

*Security.* We prove the security of  $\Pi$  in the following theorem.

**Theorem 3.1 (A Unified Framework for VSS Schemes).** *If the commitment scheme  $\mathcal{C}((\cdot, \cdot), \cdot)$  is perfectly (resp. computationally) hiding and computationally (resp. perfectly) binding, and  $\mathcal{H}$  is a (classic/quantum) random oracle, then the generic construction given in Fig. 2 constitutes a secure VSS scheme in the honest-majority setting. Specifically: (i) the Reconstruct protocol always yields a unique secret distributed by the dealer for any qualified set of shareholders, (ii) any non-qualified set of computationally unbounded (resp. bounded) shareholders cannot gain any information about the secret.*

*Proof.* The proof of this theorem can be regarded as an extension of the proof found in [19, Theorem 1]. Furthermore, we demonstrate that our proposed construction can also satisfy the *verifiable secret and polynomial sharing* property. On another front, this proof can also be seen as a slightly simplified version of the proof in [1, Theorem 3.1], where the commitments are merged and in some instantiations the quantum RO is relaxed to classic RO. It's worth noting that following a valid sharing phase, any coalition of  $t + 1$  honest parties is capable of reconstructing the witness polynomial  $f(X)$ .

As mentioned in the **Verify** algorithm, since the disqualification decision is solely based on public (broadcast) information, all honest shareholders ultimately reach the same decision. Moreover, if the dealer will be honest and follow the **Share** algorithm, then the **Verify** algorithm will return **true**, and all the honest shareholders will get a valid and distinct share of a *unique* secret.

*Verifiability.* In the honest-majority scenario where  $t+1$  of the parties are honest, with  $n \geq 2t + 1$ , this property is achieved via the random oracle  $\mathcal{H}$  and the binding property of the commitment scheme  $\mathcal{C}$ . Assume w.l.o.g. that at least

**Initial:** Parties  $P_1, \dots, P_n$  generate parameters for  $\mathcal{C}$  and each one registers a PK to facilitate secure communications. We presume the existence of a dealer  $D$  and  $P_1, \dots, P_n$  parties who receive the shares at the end of sharing phase.

**Share:** Given  $(n, t)$ , random oracle  $\mathcal{H}$ , to share  $f_0$ , the dealer  $D$  proceeds as follows:

1. Sample a uniformly random polynomial  $f(X)$  and  $r(X)$  of degree  $t$  with coefficients in a ring  $\mathbb{Z}_N$  (or a field  $\mathbb{Z}_q$ ), subject to  $f(0) = f_0$ .
2. For  $i = 1, 2, \dots, n$ : set  $f_i := f(i)$ , and  $r_i := r(i)$ .
3. For  $i = 1, 2, \dots, n$ : set  $c_i = \mathcal{C}(f_i, r_i)$  (or set  $c_i = \mathcal{C}((f_i, r_i), \gamma_i)$ , where  $\gamma_i$  are additional randomnesses used in the commitment phase).
4. Set  $z(X) = r(X) + d \cdot f(X)$  and  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , where  $d$  is the challenge value obtained from the random oracle, i.e.,  $d := \mathcal{H}(c_1, \dots, c_n)$ ;
5. Send share  $f_i$  (and  $\gamma_i$  if applicable) securely to  $P_i$  and broadcast  $\pi_{share}$ .

**Verify:** Given  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , and the individual shares:

1. Each party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, and then computes  $d := \mathcal{H}(c_1, \dots, c_n)$  and uses his/her share  $f_i$  (and  $\gamma_i$  if applicable) and checks if  $c_i = \mathcal{C}(f_i, z(i) - d \cdot f_i)$  (or  $c_i = \mathcal{C}((f_i, z(i) - d \cdot f_i), \gamma_i)$ ). If the verification of  $P_i$  fails, then  $P_i$  broadcasts a complaint against the dealer.
2. If the number of shareholders complaining against the dealer exceeds a threshold value  $t$ , the dealer will be disqualified, and the verification process will result in a **false** outcome.
3. In case a shareholder  $P_i$  raises a complaint about the verification of their part, the dealer will broadcast  $f_i = f(i)$  (and  $\gamma_i$  if applicable) to enable everyone to verify it using the verification equation. If the verification passes, the protocol continues as usual. However, if it fails, the dealer will be disqualified, leading to a **false** verification outcome. Since the disqualification decision is solely based on the information broadcasted, all honest shareholders will ultimately reach a consensus either on a set of qualified parties  $Q \subseteq \{P_1, P_2, \dots, P_n\}$  or on rejecting the final verification.

**Reconstruct:** Given  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , each party  $P_i$  broadcasts the secret value  $f_i$  (and  $\gamma_i$  if applicable). A party  $P_i$  is said to be confirmed if  $c_i = \mathcal{C}(f_i, z(i) - d \cdot f_i)$  (or  $c_i = \mathcal{C}((f_i, z(i) - d \cdot f_i), \gamma_i)$ , where  $d := \mathcal{H}(c_1, \dots, c_n)$ ). Consider  $f_i$  values of any  $t + 1$  confirmed parties and interpolate  $f(X)$  of degree  $t$  that passes through those points. Finally, the output is  $f_0 = f(0)$  or **false** (if  $t + 1$  valid shares were not obtained).

**Fig. 2. II:** A Unified Framework for Building VSS Schemes. In the general construction,  $\mathcal{H}$  represents an instantiation for the quantum/classic random oracle, and  $\mathcal{C}((\cdot, \cdot), \cdot)$  is a perfectly (resp. computationally) hiding and computationally (resp. perfectly) binding (vector) commitment scheme.

$P_1, \dots, P_{t+1}$  parties are honest. Let  $f(X)$ ,  $r(X)$  and  $z(X) := r(X) + d \cdot f(X)$  be the polynomials of degree  $t$  determined by values  $f_i$ ,  $r_i$ , and  $d \cdot f_i + r_i$ , for  $1 \leq i \leq t+1$ , where  $d$  is obtained from the random oracle, i.e.,  $d = \mathcal{H}(c_1, \dots, c_n)$ . If  $c_i = \mathcal{C}(f_i, z(i) - d \cdot f(i), \gamma_i)$  for all  $i = 1, \dots, t+1$ , then define  $f_i := f(i)$ . Otherwise, set  $f_i := 0$ .

The dealer has committed (in a distributed fashion) himself to the distinct values  $c_1, \dots, c_n$  by broadcasting  $\pi_{share} := (c_1, \dots, c_n, z(X))$  and sending  $\{f_i, \gamma_i\}_{i=1}^n$  to  $n \geq 2t + 1$  parties, where at least  $t + 1$  of them are honest. Therefore, the values  $f_i, \gamma_i$  and  $r_i := z(i) - d \cdot f_i$  for  $1 \leq i \leq t + 1$  are

set at the end of the sharing phase, and consequently the polynomial  $f(x)$  is set. Then, the value of  $f_0$  is well-defined at the end of the sharing phase, and given a degree- $t$  polynomial  $z(X)$ , and opening of  $t + 1$  commitments with points  $r_i := z(i) - d \cdot f_i$ , enables the reconstruction of degree- $t$  polynomials  $r(X) := z(X) - d \cdot f(X)$  and  $f(X)$  using Lagrange interpolation. Consequently, any  $t + 1$  honest parties will be able to collectively reconstruct the secret  $f(0)$ . Looking from a different perspective, one may notice that the general construction uses a special sound sigma protocol with designated verification. Given two acceptable transcripts  $(c_1, \dots, c_n, d, z(X))$  and  $(c_1, \dots, c_n, d' \neq d, z'(X))$ , obtained by rewiring the prover (i.e., the dealer), from the verification equation, we can write  $\mathcal{C}(f_i, z(i) - d \cdot f_i, \gamma_i) = \mathcal{C}(f_i, z'(i) - d' \cdot f_i, \gamma_i)$  for  $i = 1, \dots, n$ , where  $n \geq 2t + 1$ . Then, relying on the (computational or perfect) binding property of the commitment scheme  $\mathcal{C}$ , we can conclude that  $z(i) - d \cdot f_i = z'(i) - d' \cdot f_i$ , and therefore  $f_i = \frac{z(i) - z'(i)}{d - d'}$  for  $i = 1, \dots, n$ . Assuming that  $d - d'$  is invertible modulo  $N$  (or  $q$ ), given any set of  $t + 1$  valid (or honest) shares, an extractor can extract a *unique* degree- $t$  polynomial from the dealer.

As a result, similarly, at the end of the **Reconstruct** phase, any set of  $t + 1$  honest parties can reconstruct a *unique* degree- $t$  polynomial and output a unique value  $f_0$ . In case  $\mathcal{C}$  is a computationally binding scheme, assume by contradiction that they reconstruct  $f'_0 \neq f_0$  by choosing  $t + 1$  values  $f'_1, \dots, f'_{t+1}$ , such that  $c_i = \mathcal{C}(f'_i, z(i) - d \cdot f'_i, \gamma_i)$ . This means that the  $t$ -degree polynomials  $f'(X)$ , and  $r'(X)$  interpolated by the points  $f'_i$  and  $z(i) - d \cdot f'_i$  (resp.) have the property that  $\mathcal{C}(f'_i, z(i) - d \cdot f'_i, \gamma_i) = \mathcal{C}(f'_i, r'(i), \gamma_i) = c_i$  for  $i = 1, \dots, t + 1$ , but  $f'(X) \neq f(X)$  (as they differ in the free term), thus there must be an index  $j$  such that  $f'(j) \neq f(j)$ . Since each degree- $t$  polynomial gets unique with its  $t + 1$  distinct evaluations, then the values  $(z(j) - d \cdot f'(j))$  and  $(z(j) - d \cdot f(j))$  are a double opening for the commitment scheme  $\mathcal{C}$ , which is known to either the dealer or  $P_j$ , which contradicts the hypothesis (the binding property of  $\mathcal{C}$ ).

*Simulatability.* If the dealer is honest in the sharing phase, then the adversary sees  $t$  points on a polynomial of degree  $t$  (i.e.,  $f(X)$ ) plus a masked degree- $t$  polynomial  $z(X) := r(X) + d \cdot f(X)$  and all the commitment values  $c_i := \mathcal{C}(f_i, r_i, \gamma_i)$  for  $i = 1, \dots, n$ . But as we assume that given  $d$  and  $z_i = r_i + d \cdot f_i$  obtaining (random values)  $f_i$  and  $r_i$  from  $c_i$  is hard (or infeasible in some cases), then from commitments  $\{c_i\}_{i=1}^n$  and the masked degree- $d$  polynomial  $z(X)$ , obtaining the values of  $f(X)$  or  $r(X)$  in other points is computationally hard (or infeasible). Note that  $t$  evaluations of a degree- $t$  polynomial, information theoretically does not reveal any information about the target polynomial. Hence, the adversary cannot recover (or learn any information about) other points, including the secret value  $f(0)$ , from  $(c_1, \dots, c_n, d, z(X))$ . In other words, given the individual shares (i.e., statements) of  $t$  (corrupted) parties, it is possible to simulate the view of the adversary. To this end, w.l.o.g., given the shares  $\{f_i, \gamma_i\}_{i=1}^t$ , the simulator samples two random degree- $t$  polynomials  $r'(X)$  and  $f'(X)$ , such that  $f'(i) = f_i$  for  $i = 1, \dots, t$ . It also samples random values  $\{\gamma_i\}_{i=t+1}^n$  from the randomizer space of the commitment scheme  $\mathcal{C}$ . Then, the simulator sets  $c'_i = \mathcal{C}(f'(i), r'(i), \gamma_i)$



for  $i = 1, \dots, n$  and  $z'(X) := r'(X) + d \cdot f'(X)$ , where  $d = \mathcal{H}(c'_1, \dots, c'_n)$ . At the end, the simulator returns  $(c'_1, \dots, c'_n, d, z'(X))$  as the simulated transcript. Since the commitments  $c'_i$  are perfectly (resp. computationally) hiding, and there are unlimited number of degree- $t$  polynomials  $z'(X)$  which passes through the points  $\{f_i, \gamma_i\}_{i=1}^t$ , therefore the simulation is perfect (resp. computational) and a computationally unbounded (resp. bounded) adversary learns nothing from the transcript of the protocol about the shared secret. It is worth noting that if the dealer commits to  $(f_i, r_i)$  without the randomizer  $\gamma_i$ , the protocol transcript cannot be simulated, and the protocol achieves only unpredictability. This ensures that a computationally bounded adversary cannot recover the secret.  $\square$

*Efficiency.* As in Shamir secret sharing, the process of sharing  $f_0$  among  $n$  parties with a threshold of  $t$  requires the dealer to compute  $n$  evaluations of a degree- $t$  polynomial  $f(X)$ . Subsequently, to generate  $\pi_{share}$ , the dealer needs to compute an additional set of  $n$  evaluations for  $r(X)$ . This process also involves generating  $n$  commitments and performing  $t$  subtractions between the coefficients of  $f(X)$  and  $r(X)$ , and a single query to the random oracle  $\mathcal{H}$ , which should ideally be highly efficient in practice. During the verification phase, parties take part in the verification of  $\Pi$  (outlined in Fig. 2) and disseminate the final output to the network. As part of this procedure, each party needs to evaluate a degree- $t$  polynomial  $z(X)$  and compute a query to the random oracle and a single commitment. Regarding communication, the dealer broadcasts  $(c_1, \dots, c_n, z(X))$ , which consists of  $n$  commitments and  $t + 1$  polynomial coefficients. The dealer also securely sends a share to each participant.

## 4 Constructing VSS Schemes Via $\Pi$

The strength of  $\Pi$  lies in its generality, simplicity, and efficiency, as it only requires a secure commitment scheme  $\mathcal{C}$  and a (quantum or classic) random oracle  $\mathcal{H}$ . Commitment schemes are one of the core primitives in cryptography, and can be built efficiently. A true random oracle might not exist in real life, but with some estimations, they usually are built using cryptographic hash functions.

In this section, we employ different commitment schemes for  $\mathcal{C}$  and utilize  $\Pi$  to build several VSS schemes. The proposed schemes exhibit various trade-offs in terms of efficiency and security. To achieve this goal, we commence by reviewing established constructions from the existing literature. Subsequently, leveraging  $\Pi$ , we introduce an alternative scheme for each VSS scheme.

### 4.1 $\Pi_F$ : A Novel VSS Scheme from Pedersen Commitment

**Overview of Feldman VSS Scheme.** One of the well-known computationally secure VSS protocols is Feldman's scheme, which is based on Shamir and was proposed by Feldman in [16]. In Feldman's scheme, given  $(n, t)$  and group generator  $g_1$ , to share a *high-entropy* secret  $f_0$ , the dealer proceeds as follows:

1. Sample a uniformly random degree- $t$  polynomial  $f(X) := f_0 + a_1X + \dots + a_tX^t$  with coefficients in  $\mathbb{Z}_q$ , subject to  $f(0) = f_0$ .

2. For  $i = 1, 2, \dots, n$ : set  $f_i := f(i)$ .
3. Compute  $c_0 = g_1^{f_0}$  and  $c_j = g_1^{a_j}$  for  $j = 1, 2, \dots, t$ .
4. Set  $\pi_{share} := (c_0, c_1, \dots, c_t)$ ; Sends share  $f_i$  securely to party  $P_i$  and broadcast  $\pi_{share}$  as the proof.

Then, to verify their received shares, given  $\pi_{share} := (c_0, c_1, \dots, c_t)$ , and the individual shares  $\{f_i\}_{i=1}^n$ : each party  $P_i$  uses his/her share  $f_i$  and checks if  $g_1^{f_i} = \prod_{j=0}^t c_j^{i^j}$  and outputs either **true** or **false**. For  $n \geq 2t + 1$ , if *all*  $n$  parties return **true**, then the final **Verify** will return **true**. Otherwise, any possible conflict between the dealer and the parties will be solved using a known conflict resolution approach (also used in **II**).

**$\Pi_F$ : An Efficient Alternative to Feldman Scheme.** By instantiating **II** with a Pedersen commitment [23], with two random group generators  $(g_1, g_2) \in \mathbb{G}$ , i.e. by setting  $c_i := g_1^{f_i} g_2^{r_i}$ , we obtain a novel VSS scheme, referred to as  **$\Pi_F$** . This scheme provides an alternative construction to the Feldman scheme [16]. In Fig. 3, we provide a concise overview of  **$\Pi_F$** , focusing solely on the steps that deviate from our general construction **II**.

**Share:** Given two random group generators  $g_1$  and  $g_2$ , the parameters  $n$  and  $t$ , to share  $f_0$ , the dealer follows the steps outlined in Fig. 2, specifically, with the following deviations:

3. For  $i = 1, 2, \dots, n$ : Compute  $c_i = g_1^{f_i} g_2^{r_i}$ .

**Verify:** Given  $g_1, g_2, \pi_{share} := (c_1, \dots, c_n, z(X))$ , and the shares  $\{f_i\}_{i=1}^n$ :

1. Each party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, and then computes  $d = \mathcal{H}(c_1, \dots, c_n)$  and uses his/her share  $f_i$  and checks if  $c_i = g_1^{f_i} g_2^{z(i) - d \cdot f_i}$ . If the verification of  $P_i$  fails, then  $P_i$  broadcasts a complaint against the dealer.

**Reconstruct:** Using the reconstruction approach outlined in Fig. 2, given  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , each party  $P_i$  broadcasts the secret value  $f_i$ . A party  $P_i$  is said to be confirmed if  $c_i = g_1^{f_i} g_2^{z(i) - d \cdot f_i}$ , where  $d := \mathcal{H}(c_1, \dots, c_n)$ . Consider  $f_i$  values of any  $t + 1$  confirmed parties and interpolate  $f(X)$  of degree  $t$  that passes through those points. Finally, the output is  $f_0 = f(0)$  or **false** (if  $t + 1$  valid shares were not obtained).

**Fig. 3.  $\Pi_F$ :** A novel VSS scheme based on discrete logarithm.

Under DL assumption, Theorem 3.1 and its security proof can be adapted for  **$\Pi_F$** , showing that it can satisfies *Verifiability* and *Unpredictability*, defined in Sec. 3.1. For completeness, we present the details in App. A. Note that, as in the Feldman scheme, in  **$\Pi_F$** ,  $f_0$  should contain sufficient entropy, and the new VSS scheme is, at best, secure against computationally bounded (classical) adversaries. In terms of efficiency,  **$\Pi_F$**  can have considerably faster verification and reconstruction compared to the Feldman scheme. However, this advantage comes at the cost of approximately  $2.5\text{-}3 \times$  slower sharing and  $2 \times$  increased communication. Please refer Tab. 1 for the details.

We highlight that it is possible to modify the general construction of **II** to work with discrete logarithms (as in Feldman's VSS) in  **$\Pi_F$** , e.g., by setting

$c_i := g^{f_i}$  and  $c'_i = g^{r_i}$ . In this case, the verification check would involve verifying whether  $c_i = g^{f_i}$  and  $c'_i = g^{z(i)-df_i}$ . However, as can be observed, these modifications would result in less efficient schemes in terms of efficiency.

## 4.2 $\Pi_P$ : A Novel VSS Scheme from Pedersen Commitment

**Overview of Pedersen VSS Scheme.** The Pedersen VSS scheme is a variation of Feldman's scheme [16] that uses a perfectly hiding commitment scheme. In Pedersen VSS scheme, the commitment takes the form of a Pedersen commitment, denoted as  $c_i = g_1^{a_i} g_2^{b_i}$ , where  $a_i, b_i$  are the coefficients of two degree- $t$  polynomials. This approach ensures that fewer than  $t$  parties receive no information about the secret, thereby achieving perfect (or IT) simulatability. In the Pedersen scheme, given two random group generators  $g_1$  and  $g_2$ , and parameters  $n$  and  $t$ , the process of sharing  $f_0$  is done as follows:

1. Sample two random degree- $t$  polynomials  $f(X) := f_0 + a_1X + \dots + a_tX^t$  and  $r(X) := r_0 + b_1X + \dots + b_tX^t$  with coefficients in  $\mathbb{Z}_q$ , subject to  $f(0) = f_0$ .
2. For  $i = 1, 2, \dots, n$ : set  $f_i := f(i)$  and  $r_i := r(i)$ .
3. Compute  $c_0 = g_1^{f_0} g_2^{r_0}$  and  $c_j = g_1^{a_j} g_2^{b_j}$  for  $j = 1, 2, \dots, t$ .
4. Set  $\pi_{share} := (c_0, c_1, \dots, c_t)$ ; Sends share  $(f_i, r_i)$  securely to party  $P_i$  and broadcast  $\pi_{share}$  as the proof.

Then, to verify their received shares, given  $\pi_{share} := (c_0, c_1, \dots, c_t)$ , and the individual shares  $\{f_i, r_i\}_{i=1}^n$ : each party  $P_i$  uses his/her share  $(f_i, r_i)$  and checks if  $g_1^{f_i} g_2^{r_i} = \prod_{j=0}^t c_j^{i^j}$  and outputs either **true** or **false**. The rest of the verification is similar to the Feldman scheme.

**$\Pi_P$ : An Efficient Alternative to Pedersen Scheme.** Instantiating  $\Pi$  with an extended (or vector) variant of the Pedersen commitment scheme from [8], specifically by employing *three* randomly chosen group generators  $(g_1, g_2, g_3) \in \mathbb{G}$ , leads to the construction of a novel DL-based VSS scheme denoted as  $\Pi_P$ . This scheme can be considered as an alternative to the Pedersen VSS scheme. In  $\Pi_P$ , commitments  $c_i$  are computed as  $c_i = g_1^{f_i} g_2^{r_i} g_3^{\gamma_i}$  for  $i = 1, \dots, n$ , where  $\gamma_i \in \mathbb{Z}_q$  are additional randomnesses used in the commitment phase. The purpose of  $\gamma_i$  is to achieve perfect (a.k.a., IT) simulatability. Thus, in order to ensure its effect is not canceled, before computing the value  $z(X) = r(X) + d \cdot f(X)$ , the dealer additionally checks if  $g_1 \neq g_2^d$  and continues if the check passes. This check is necessary to ensure that the simulation of the transcript is perfect, and the protocol satisfies IT simulatability against up to  $t$  shareholders. At the end, along with the share  $f_i$ , the dealer also sends the randomizer  $\gamma_i$  to party  $P_i$ . Then, given public values  $(g_1, g_2, g_3, c_i, z(X))$  and secret values  $(f_i, \gamma_i)$ , party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, and then computes  $d = \mathcal{H}(c_1, \dots, c_n)$  and verifies if  $c_i = g_1^{f_i} g_2^{z(i)-d \cdot f_i} g_3^{\gamma_i}$  and outputs either **true** or **false**. The full description of  $\Pi_P$  is summarized in Fig. 4.

Under DL assumption, Theorem 3.1 and its security proof can be adapted for  $\Pi_P$ . Notably in this case, since the commitment to  $(f_i, r_i)$  is perfectly hiding and

**Share:** Given three random group generators  $(g_1, g_2, g_3)$ , the parameters  $(n, t)$ , to share  $f_0$ , the dealer follows the steps outlined in Fig. 2, but, with the following deviations:

3. For  $i = 1, 2, \dots, n$ , sample random values  $\gamma_i \in \mathbb{Z}_q$  and set  $c_i = g_1^{f_i} g_2^{r_i} g_3^{\gamma_i}$ .
4. Compute the challenge value  $d := \mathcal{H}(c_1, \dots, c_n)$ ; Check if  $g_1 \neq g_2^d$  and if the check passed, set  $z(X) = r(X) + d \cdot f(X)$  and  $\pi_{share} := (c_1, \dots, c_n, z(X))$ . Otherwise, restart the protocol from step 3;
5. Send shares  $(f_i, \gamma_i)$  securely to party  $P_i$  and broadcast  $\pi_{share}$  as the proof.

**Verify:** Given random group generators  $(g_1, g_2, g_3)$ , a proof  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , and the shares  $\{f_i, \gamma_i\}_{i=1}^n$ :

1. Each party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, computes challenge value  $d = \mathcal{H}(c_1, \dots, c_n)$  and then uses his/her shares  $(f_i, \gamma_i)$  and checks if  $c_i = g_1^{f_i} g_2^{z(i)-d \cdot f_i} g_3^{\gamma_i}$ . If the verification of  $P_i$  fails, then  $P_i$  broadcasts a complaint against the dealer.

**Reconstruct:** Using the reconstruction approach outlined in Fig. 2, given  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , each party  $P_i$  broadcasts the secret values  $f_i$  and  $\gamma_i$ . A party  $P_i$  is said to be confirmed if  $c_i = g_1^{f_i} g_2^{z(i)-d \cdot f_i} g_3^{\gamma_i}$ , where  $d := \mathcal{H}(c_1, \dots, c_n)$ . Consider  $f_i$  values of any  $t+1$  confirmed parties and interpolate  $f(X)$  of degree  $t$  that passes through those points. Finally, the output is  $f_0 = f(0)$  or **false** (if  $t+1$  valid shares were not obtained).

**Fig. 4.  $\Pi_P$ :** A novel IT-secure VSS scheme from Pedersen commitments.

$g_1 \neq g_2^d$ , the simulation of transcript is perfect and the resulting VSS scheme can achieve IT simulatability, and a computationally unbounded adversary  $\mathcal{A}$  who controls up to  $t$  parties, cannot learn anything about the other shares and the main secret from the transcript of the protocol, i.e.,  $(c_1, \dots, c_n, d, z(X))$ . Note that in this case, there is a negligible probability that the dealer may need to repeat step 3 in the sharing phase. The resulting VSS scheme  $\Pi_P$  surpasses Pedersen VSS scheme [23] in the verification and reconstruction phases. Please refer to Tab. 1 for detailed comparisons.

We highlight that, it is possible to modify  $\Pi$  to work with the original Pedersen commitment (i.e., with two generators, instead of three) in  $\Pi_P$ , e.g., by setting  $c_i := g^{f_i} h^{\gamma_i}$  and  $c'_i = g^{r_i} h^{\gamma_i}$  and updating the verification equation accordingly. However, these modifications would result in less efficient variant of  $\Pi_P$  in terms of both (public and private) communications and computation.

### 4.3 $\Pi_{LA}$ : A Novel VSS Scheme from Hash Functions

*Overview of the RO-based VSS Scheme of ABCP.* Recently, Atapoor, Bagheri, Cozzo, and Pedersen [1], proposed a general construction and showed that given a NI-TZK proof scheme for the Shamir relation, given in eq. (1), one can build a VSS scheme based on Shamir secret sharing. Following their initial result, they built a NI-TZK proof scheme for the Shamir relation, and then used it to construct a novel PQ-secure VSS scheme. Their resulting VSS scheme uses NI-TZK proofs, which use a quantum RO (qRO) and a quantum computationally hiding commitment scheme, which both are built from hash functions. Their

construction is extremely efficient and outperforms the prior computationally secure VSS schemes from the literature. In their scheme, given  $n$  and  $t$ , to share the secret  $f_0$ , the dealer proceeds as follows:

1. Sample two random degree- $t$  polynomials  $r(X) := r_0 + b_1X + \dots + b_tX^t$  and  $f(X) := f_0 + a_1X + \dots + a_tX^t$  with coefficients in  $\mathbb{Z}_N$ , subject to  $f(0) = f_0$ .
2. For  $i = 1, 2, \dots, n$ : set  $f_i := f(i)$  and  $r_i := r(i)$ , and also samples two vectors of randomnesses  $y_i, y'_i$ .
3. Compute  $c_i = \mathcal{C}(f(i), y_i)$  and  $c'_i = \mathcal{C}(r(i), y'_i)$  for  $i = 1, 2, \dots, n$ , where  $\mathcal{C}$  is a quantum computationally hiding commitment scheme.
4. Set the challenge value  $d = \mathcal{H}(c_1, \dots, c_n, c'_1, \dots, c'_n)$ , where  $\mathcal{H}$  is a qRO.
5. Set the response  $z(X) = r(X) - d \cdot f(X)$ ;
6. Finally, set  $\pi_{share} := (c_1, \dots, c_n, c'_1, \dots, c'_n, z(X))$ ; Sends share  $f_i$  and the randomnesses  $(y_i, y'_i)$  securely to party  $P_i$  and broadcast  $\pi_{share}$  as the proof.

*Verify.* To verify their received shares, given  $\pi_{share} := (c_1, \dots, c_n, c'_1, \dots, c'_n, z(X))$ , and the individual shares  $\{f_i\}_{i=1}^n$  and randomnesses  $\{y_i, y'_i\}_{i=1}^n$ : each party  $P_i$  uses his/her share  $(f_i, y_i, y'_i)$  and proceeds as follows: 1) checks if  $\mathcal{C}(f_i, y_i) = c_i$ ; 2) computes the challenge value  $d = \mathcal{H}(c_1, \dots, c_n, c'_1, \dots, c'_n)$ ; 3) checks if  $\mathcal{C}(z(i) + df_i, y'_i) = c'_i$ ; and outputs either **true** or **false**. The rest of the verification, i.e., conflict resolution, is the same as in  **$\Pi$**  (given in Fig. 2).

**$\Pi_{LA}$** : *More Efficient VSS Scheme from Hash Functions.* By instantiating  **$\Pi$**  with a non-homomorphic commitment scheme, like those based on hash functions, we obtain a novel PQ-secure VSS scheme in the quantum random oracle model, named  **$\Pi_{LA}$** . In  **$\Pi_{LA}$** , in case  $f_0$  has enough entropy, commitments  $c_i$  can be computed as  $c_i = \mathcal{H}(f_i, r_i)$  for  $i = 1, \dots, n$ , where  $\mathcal{H}$  is a well-defined secure hash function and the coefficients of  $f(X)$  and  $r(X)$  are sampled randomly from ring  $\mathbb{Z}_N$ <sup>4</sup>. On the other side, given  $(f_i, c_1, \dots, c_n, z(X))$ , party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, and then computes  $d = \mathcal{H}(c_1, \dots, c_n)$  and verifies if  $c_i = \mathcal{H}(f_i, z(i) - d \cdot f_i)$  and outputs either **true** or **false**. In the general case, if  $f_0$  lacks enough entropy, the dealer can follow the approach in Fig. 4 and use an additional randomizer  $\gamma_i$  in the commitments, i.e.,  $c_i = \mathcal{H}(f_i, r_i, \gamma_i)$  for  $i = 1, \dots, n$ . Accordingly, to verify their shares, for the last check, party  $P_i$  uses  $(f_i, \gamma_i)$  and verifies if  $c_i = \mathcal{H}(f_i, z(i) - d \cdot f_i, \gamma_i)$  and outputs either **true** or **false**. The description of  **$\Pi_{LA}$**  is summarized in Fig. 5.

Under the quantum security of commitment scheme  $\mathcal{H}$  and the quantum random oracle  $\mathcal{H}$ , Theorem 3.1 and its security proof can be adapted for the general case of  **$\Pi_{LA}$** . The proof, can also be written similar to the proof of Theorem 2 in [1]. Note that in this case  $c_i$  hides  $f_i, r_i$  against computationally

<sup>4</sup> Note that, in this case the challenge value  $d$  is sampled from an exceptional set. Therefore, when  $\mathbb{Z}_N$  is a cryptographically sized field, we can achieve a negligible error rate in  **$\Pi_{LA}$** , i.e. below  $2^{-\lambda}$ . In cases where  $\mathbb{Z}_N$  is a ring, it's possible to encounter situations where the largest exceptional set has a size of  $k < 2^\lambda$ . In such scenarios, the dealer needs to amplify the soundness error rate in a standard manner by repeating the protocol  $l = \lceil \lambda / \log k \rceil$  times.

**Share:** Given a secure hash function  $H$ , quantum random oracle  $\mathcal{H}$ , the parameters  $n$  and  $t$ , to share  $f_0$ , the dealer follows the steps outlined in Fig. 2, but, with the following deviations:

- For  $i = 1, 2, \dots, n$ : Compute commitments  $c_i = H(f_i, r_i)$  (or set  $c_i = H(f_i, r_i, \gamma_i)$  if  $f_0$  lacks enough entropy, where  $\gamma_i$  are sampled randomly).

**Verify:** Given  $H$  and  $\mathcal{H}$ ,  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , and the shares  $\{f_i\}_{i=1}^n$ :

- Each party  $P_i$  first checks if  $z(X)$  is a degree  $t$  polynomial, and if so, computes  $d = \mathcal{H}(c_1, \dots, c_n)$  and then uses his/her share  $f_i$  (and  $\gamma_i$ ) and checks if  $c_i = H(f_i, z(i) - d \cdot f_i)$  (or  $c_i = H((f_i, z(i) - d \cdot f_i), \gamma_i)$ ). If the verification of  $P_i$  fails, then  $P_i$  broadcasts a complaint against the dealer.

**Reconstruct:** Using the reconstruction approach outlined in Fig. 2, given  $\pi_{share} := (c_1, \dots, c_n, z(X))$ , each party  $P_i$  broadcasts the secret value  $f_i$  (and  $\gamma_i$  if applicable). A party  $P_i$  is said to be confirmed if  $c_i = H(f_i, z(i) - d \cdot f_i)$  (or  $c_i = H((f_i, z(i) - d \cdot f_i), \gamma_i)$ ), where  $d := \mathcal{H}(c_1, \dots, c_n)$ . Consider  $f_i$  values of any  $t + 1$  confirmed parties and interpolate  $f(X)$  of degree  $t$  that passes through those points. Finally, the output is  $f_0 = f(0)$  or **false** (if  $t + 1$  valid shares were not obtained).

**Fig. 5.  $\Pi_{LA}$ :** A novel PQ-secure VSS scheme from hash functions.

bounded (quantum) adversaries.  $\Pi_{LA}$  outperforms the ABCP scheme [1], by a constant factor in terms of all efficiency metrics (please refer to Tab. 1, and 2).

#### 4.4 Efficiency Comparisons of New VSS Schemes

We conducted an analysis of the asymptotic costs for the proposed VSS schemes ( $\Pi_F, \Pi_P, \Pi_{LA}$ ) and compared them to relevant constructions from existing literature. A summary of the results can be found in Table 1.

**Empirical Performance of Pedersen,  $\Pi_P$ , ABCP, and  $\Pi_{LA}$  Schemes.** In addition, we assessed the practical performance of  $\Pi_P$  and  $\Pi_{LA}$  through a prototype implementation in SageMath and compared their performance with the Pedersen scheme and the recently proposed ABCP construction [1]<sup>5</sup>. We used the source code implementations for the Pedersen and ABCP schemes from [1]<sup>6</sup>. Our experiments are done using the elliptic curve Ed25519 and the hash function SHA256 for both commitment and random oracle, on a laptop with Ubuntu 22.04 LTS, a 11th Gen Intel(R) Core(TM) i9-11950H at base frequency 2.60GHz, and 128GB of memory. Both the sharing and verification algorithms are executed in single-thread mode. We have summarized our implementation results for various parameter sets in Tab. 2.

Upon comparing the implementation outcomes of the Pedersen scheme with those of  $\Pi_P$ , it becomes apparent that  $\Pi_P$  yields a remarkable acceleration in the verification phase. Notably,  $\Pi_P$  achieves verification times that are 12 $\times$ , 82 $\times$ , and 105 $\times$  faster in comparison to the Pedersen scheme for the parameter pairs  $(n, t)$  equal to (128, 63), (2048, 1023), and (16384, 8191), respectively. In terms

<sup>5</sup> Our source code is publicly available at <https://github.com/Bagheri/VSS-Pi>.

<sup>6</sup> Available on <https://github.com/Bagheri/VSS-ABCP23>.

**Table 2.** Implementation results of VSS schemes Pedersen [23],  $\Pi_P$ , ABCP [1], and  $\Pi_{LA}$ .  $n$ : Number of parties,  $t$ : Threshold value,  $|\mathbb{Z}_q| = |\mathbb{Z}_N| = |\mathbb{G}| = |\mathcal{H}| = 256$  bits.

| $(n, t)$      | Metrics               | Pedersen [23] | $\Pi_P$    | ABCP [1]  | $\Pi_{LA}$ |
|---------------|-----------------------|---------------|------------|-----------|------------|
| (32, 15)      | Share                 | 74.8 msec     | 222.1 msec | 2.2 msec  | 1.7 msec   |
|               | Verify                | 10.7 msec     | 7.1 msec   | 0.13 msec | 0.10 msec  |
|               | Dealer's Broadcast    | 0.5 KB        | 1.5 KB     | 2.5 KB    | 1.5 KB     |
|               | Dealer's Private Com. | 2.0 KB        | 2.0 KB     | 1.0 KB    | 1.0 KB     |
| (128, 63)     | Share                 | 303 msec      | 852 msec   | 13.2 msec | 11.4 msec  |
|               | Verify                | 99.2 msec     | 8.3 msec   | 0.37 msec | 0.33 msec  |
|               | Dealer's Broadcast    | 2.0 KB        | 6.0 KB     | 10.0 KB   | 6.0 KB     |
|               | Dealer's Private Com. | 8.0 KB        | 8.0 KB     | 4.0 KB    | 4.0 KB     |
| (512, 255)    | Share                 | 1.29 sec      | 3.63 sec   | 0.13 sec  | 0.12 sec   |
|               | Verify                | 552 msec      | 12.5 msec  | 1.2 msec  | 1.1 msec   |
|               | Dealer's Broadcast    | 8.0 KB        | 24.0 KB    | 40.0 KB   | 24.0 KB    |
|               | Dealer's Private Com. | 32.0 KB       | 32.0 KB    | 16.0 KB   | 16.0 KB    |
| (2048, 1023)  | Share                 | 6.45 sec      | 16.07 sec  | 1.81 sec  | 1.78 sec   |
|               | Verify                | 2.32 sec      | 28.3 msec  | 4.9 msec  | 4.8 msec   |
|               | Dealer's Broadcast    | 32.0 KB       | 96.0 KB    | 160.0 KB  | 96.0 KB    |
|               | Dealer's Private Com. | 128.0 KB      | 128.0 KB   | 64.0 KB   | 64.0 KB    |
| (8192, 4095)  | Share                 | 47.1 sec      | 84.6 sec   | 28.6 sec  | 28.5 sec   |
|               | Verify                | 9.38 sec      | 0.092 sec  | 0.020 sec | 0.018 sec  |
|               | Dealer's Broadcast    | 128 KB        | 384 KB     | 640 KB    | 384 KB     |
|               | Dealer's Private Com. | 512 KB        | 512 KB     | 256 KB    | 256 KB     |
| (16384, 8191) | Share                 | 149.0 sec     | 221.69 sec | 112.0 sec | 111.5 sec  |
|               | Verify                | 18.7 sec      | 0.178 sec  | 0.070 sec | 0.050 sec  |
|               | Dealer's Broadcast    | 256 KB        | 768 KB     | 1280 KB   | 768 KB     |
|               | Dealer's Private Com. | 1024 KB       | 1024 KB    | 512 KB    | 512 KB     |

of communication costs,  $\Pi_P$  demands a slightly larger data broadcast from the dealer, amounting to  $3\times$  compared to  $1\times$  in the Pedersen scheme. We highlight that these achievements within  $\Pi_P$  are accompanied by a slightly slower sharing phase, resulting in speeds ranging from  $1.48\text{--}2.96\times$  in comparison to the baseline of  $1\times$  in Pedersen scheme. Moreover, it's worth noting that the disparity in costs becomes less pronounced as the values of  $(n, t)$  increase. For instance, in the case of  $(n, t) = (16384, 8191)$ , the sharing phase of  $\Pi_P$  is approximately 48% slower than the sharing phase of the Pedersen scheme. We believe that this gap and the verification time can be reduced through various optimization techniques. One optimization approach is to use improved algorithms for evaluating polynomials at multiple points.

Regarding,  $\Pi_{LA}$ , we can see that it is slightly more efficient than the recent construction by ABCP [1] in terms of computation and communication costs. Notably, owing to their reliance on lightweight cryptography and polynomial evaluations exclusively, both  $\Pi_{LA}$  and ABCP [1] exhibit swifter performance than  $\Pi_P$  and the Pedersen scheme [23], which relies on asymmetric cryptography.

Regarding the efficiency of the reconstruction phase in all studied and new DL-based VSS schemes, as shown in Table 1, we observe that the new schemes require  $t$  times fewer exponentiations but at the cost of  $t$  polynomial evaluations.



By a rough estimation, if we multiply the verification time of each VSS scheme by  $t$ , we can obtain a rough estimation of the time required for secret reconstruction. For instance, for  $(n, t) = (16384, 8191)$ , the secret reconstruction phase for  $\Pi_P$  could be approximately  $8191 \times$  faster compared to the Pedersen VSS. We note that in practice, this gap can be narrowed by employing multi-exponentiation techniques in both schemes, and we expect it can be more effective for the Pedersen VSS scheme.

It is also worth mentioning that our implementation is done using SageMath, and it remains relatively basic, functioning as a single-threaded process without specific optimizations. Given that our proposed schemes heavily rely on polynomial evaluations, an effective optimization is to employ more efficient algorithms for the evaluation of a polynomial at multiple points, as outlined in [29]. Independently, the developers of Dock Network have recently presented an efficient Rust implementation of the  $\Pi_{LA}$  VSS scheme <sup>7</sup>.

## 5 Generalizing DL and Schnorr Over Polynomials

This section introduces an efficient NIZK PoK scheme that can serve as a tool for  $\Pi$  while maintaining relevance for other threshold schemes and applications.

Let  $\mathbb{G}$  be a group with hard DL, and  $g$  be the group generator. Let a prover aim to convince a verifier that for the public statement  $F \in \mathbb{G}$ , he knows a witness  $f \in \mathbb{Z}_q$  which holds in relation  $R_{DL} = \{(g, F), f \mid F = g^f\}$ . Schnorr's known ID protocol [24] allows a prover to efficiently achieve this goal. In the NIZK version of Schnorr's ID protocol, given  $g, f$ , a prover samples a randomness  $r \in \mathbb{Z}_q$ , sets  $\Gamma = g^r$ , and publishes  $\Gamma$  and  $z = r + d \cdot f \pmod q$  as the proof, where  $d$  is the challenge value obtained from the random oracle  $\mathcal{H}$ , i.e.,  $d := \mathcal{H}(F, \Gamma)$ . Then, given the statement  $(g, F)$  and the proof  $(\Gamma, z)$ , a verifier first sets  $d = \mathcal{H}(F, \Gamma)$ , and then checks if  $g^z = \Gamma F^d$ , and returns **true** or **false**.

Next, we generalize Schnorr's ID protocol and present a NIZK PoK scheme for the Polynomial DL (PDL) relation  $R_{PDL}$ , defined as follows,

$$R_{PDL} = \{(g, x_i, F_i), f(X) \mid F_i = g^{f(x_i)}, \text{ for } i = 1, 2, \dots, n\}, \quad (3)$$

where  $f(X) \in \mathbb{Z}_q[X]_t$  is a (at most) degree  $t \leq n - 1$  witness polynomial with coefficients defined over  $\mathbb{Z}_q$ , and  $x_1, \dots, x_n$  are  $n$  *distinct* elements from  $\mathbb{Z}_q$ . The  $R_{PDL}$  relation is base on the PDL problem defined as follows.

**Definition 5.1 (Polynomial Discrete Logarithm Problem).** *Let  $\mathbb{G}$  be a finite cyclic group of order  $q$  generated by  $g$ . Given  $F_1, \dots, F_n$  from  $\mathbb{G}$  and distinct elements  $x_1, \dots, x_n$  from  $\mathbb{Z}_q$ , find a polynomial  $f(X) \in \mathbb{Z}_q[X]_t$  of (at most) degree  $t$ , where  $0 \leq t \leq n - 1$ , such that  $F_i = g^{f(x_i)}$  for all  $i = 1, \dots, n$ .*

*In other words, an algorithm  $\mathcal{A}$  has advantage  $\epsilon$  in solving PDL in  $\mathbb{G}$  if*

$$\Pr[\mathcal{A}(x_1, \dots, x_n, g, g^{f(x_1)}, \dots, g^{f(x_n)}) = f(X)] \geq \epsilon$$

<sup>7</sup> See [https://github.com/docknetwork/crypto/blob/main/secret\\_sharing\\_and\\_dkg/src/baghery\\_vss.rs](https://github.com/docknetwork/crypto/blob/main/secret_sharing_and_dkg/src/baghery_vss.rs).



**Prover:** Given the statement  $(g, x_i, F_i)_{i=1}^n$  and the witness polynomial  $f(X)$ , proceed as follows and output a proof  $\pi$ .

1. Sample a degree- $t$  polynomial  $r(X) \in \mathbb{Z}_q[X]_t$ ; Set  $\{\Gamma_i = g^{r(x_i)}\}_{i=1}^n$ .
2. Set  $d \leftarrow \mathcal{H}(F_1, \dots, F_n, \Gamma_1, \dots, \Gamma_n)$ , where  $\mathcal{H}$  is a random oracle.
3. Set  $z(X) = r(X) + d \cdot f(X) \pmod{q}$ ;
4. Return  $\pi := (\Gamma_1, \dots, \Gamma_n, z(X))$

**Verifier:** Given statement  $(g, \{x_i, F_i\}_{i=1}^n)$  and  $\pi := (\Gamma_1, \dots, \Gamma_n, z(X))$ , the verifier first checks if  $z(X)$  is a degree- $t$  polynomial. If so, then sets  $d \leftarrow \mathcal{H}(F_1, \dots, F_n, \Gamma_1, \dots, \Gamma_n)$  and checks if:  $g^{z(x_i)} = \Gamma_i(F_i)^d$  for  $i = 1, \dots, n$ , and outputs **true** or **false**. Note that to make the communication shorter, as in Schnorr's ID protocol, alternatively, the prover could publish  $\pi := (d, z(X))$ , and then the verifier would need to check if  $z(X)$  is a degree- $t$  polynomial and  $d = \mathcal{H}(F_1, \dots, F_n, \frac{g^{z(x_1)}}{F_1^d}, \dots, \frac{g^{z(x_n)}}{F_n^d})$ .

**Fig. 6.**  $\pi_{PDL}$ : An efficient NIZK proof of knowledge for  $R_{PDL}$ .

where  $f(X) \in \mathbb{Z}_q[X]_t$  is (at most) a degree- $t$  polynomial with  $0 \leq t \leq n-1$ , and the probability is over the random choice of generator  $g \in \mathbb{G}^*$  and the distinct choice of  $x_1, \dots, x_n$  in  $\mathbb{Z}_q$ . We say that the  $(t, \epsilon)$ -PDL assumption holds in  $\mathbb{G}$  if no  $t$ -time algorithm has advantage at least  $\epsilon$  in solving the PDL problem in  $\mathbb{G}$ .

Occasionally we drop the  $t$  and  $\epsilon$  and refer to the PDL assumption in  $\mathbb{G}$ . It can be seen that the hardness of the PDL problem can be reduced to that of the DL problem. As an instance, let  $\mathcal{A}$  be an adversary against the PDL problem, and  $(g, h := g^f)$  be the challenge values for the DL problem. Then, one can construct an adversary  $\mathcal{B}$  that acts as follows.  $\mathcal{B}$  sets  $F_1 := h, x_1 := 1, x_2 := 2$ , and additionally samples another random element  $F_2$  from  $\mathbb{G}$ , and sends the tuple  $(g, x_1, x_2, F_1, F_2)$  to the adversary  $\mathcal{A}$ . If  $\mathcal{A}$  returns  $f(X)$  such that  $F_1 = g^{f(1)}$  and  $F_2 = g^{f(2)}$ , then  $\mathcal{B}$  returns  $f(1)$  as the answer to the DL challenge.

We assume  $t+1 \leq n$  in the PDL problem, however, it's worth noting that as we increase  $n$ , we add more evaluations of  $f(X)$  into the statement. Thus, we anticipate the existence of an upper bound for  $n$  (for a specific  $t$ ), and we leave it as an interesting feature research question.

**Generalization of Schnorr Protocol.** In Fig. 6, we introduce a generalized version of Schnorr's NIZK PoK protocol, which enables a prover to generate a NIZK proof of knowledge for the relation  $R_{PDL}$ , as defined in Eq. (3).

**Theorem 5.1 (A NIZK Proof of Knowledge for  $R_{PDL}$ ).** *Let  $g$  be the generator of  $\mathbb{G}$ ,  $\{F_i\}_{i=1}^n \in \mathbb{G}$ ,  $\{x_i\}_{i=1}^n$  be  $n$  distinct elements from  $\mathbb{Z}_q$ , and  $t$  be the (maximum) degree of witness polynomial  $f(X)$ . Assuming PDL is hard, for  $0 \leq t < n$ , the protocol  $\pi_{PDL}$  (described in Fig. 6) is a NIZK PoK for  $R_{PDL}$  in the RO model.*

*Proof.* We first prove the security of the interactive case, and then using standard Fiat-Shamir transform, extend it to the non-interactive case in the RO model.

*Completeness.* If the prover and verifier honestly follow the protocol, for  $i = 1, \dots, n$ , we have

$$g^{z(x_i)} = g^{r(x_i)+df(x_i)} = g^{r(x_i)} + (g^{f(x_i)})^d = \Gamma_i F_i^d.$$

*Special Soundness:* Let  $(\Gamma_i, d, z(X))$  and  $(\Gamma_i, d', z'(X))$  be two acceptable transcripts with the same commitments and different challenge values, that are obtained by rewinding. Then, from the verification equation, we know that for  $i = 1, \dots, n$ :

$$g^{z(x_i)} = \Gamma_i (F_i)^d \quad , \quad g^{z'(x_i)} = \Gamma_i (F_i)^{d'}.$$

This implies that, for  $i = 1, \dots, n$ :

$$g^{z(x_i)-z'(x_i)} = F_i^{d-d'} \Rightarrow F_i = g^{\frac{z(x_i)-z'(x_i)}{d-d'}}.$$

Since  $z(X)$  is a degree- $t$  polynomial, therefore, if all the  $n \geq t+1$  of the checks pass, from  $f_i := \frac{z(x_i)-z'(x_i)}{d-d'}$  for  $i = 1, \dots, n$ , we can obtain  $n \geq t+1$  *distinct* evaluations of a unique degree- $t$  polynomial at points  $x_1, \dots, x_n$ . Considering the fact that any degree- $t$  polynomial can be determined from its  $t+1$  *distinct* evaluations, using Lagrange interpolation, w.l.o.g. an extractor can use  $\{f_i\}_{i=1}^{t+1}$  and reconstruct (extract) a *unique* degree- $t$  polynomial  $f(X)$ , which is a witness (resp. solution) for  $R_{PDL}$  relation (resp. PDL problem).

*Honest Verifier Zero-Knowledge (HVZK):* Next, we show that given the statement  $(g, \{x_i, F_i\}_{i=1}^n)$  and the challenge value  $d$ , a simulator can simulate the transcript of the protocol. To this end, the simulator first randomly samples a degree- $t$  polynomial  $z'(X) \in \mathbb{Z}_q[X]_t$ . Then, for  $i = 1, \dots, n$ : sets  $\Gamma'_i = \frac{g^{z'(x_i)}}{F_i^d}$ . Finally, the simulator returns  $(\{\Gamma'_i\}_{i=1}^n, d, z'(X))$  as the simulated proof. As it can be seen, since  $z'(X)$  is sampled randomly, therefore  $\{\Gamma'_i\}_{i=1}^n$  are also random, and the simulated proof is indistinguishable from the real one.

Since the interactive scheme is public coin, and satisfies completeness, (perfect) special soundness, and (computational) HVZK, then, in the random oracle model, using Fiat-Shamir transform [17], it can be turn into a NIZK proof of knowledge scheme for  $R_{PDL}$  (defined in eq. (3)).  $\square$

**Efficiency of  $\pi_{PDL}$  and Related works.** In  $\pi_{PDL}$ , a prover needs to evaluate a degree- $t$  polynomial in  $n$  points, and compute  $n$  EXP in  $\mathbb{G}$  and a single hash. Subsequently, the prover publishes a proof  $\pi := (d, z(X))$ , comprising  $t+2$  field elements. On the other side, a verifier needs to evaluate a degree- $t$  polynomial in  $n$  points, and compute  $2n$  EXP, and 1 hashing operation.

In [6, 12], authors introduced two different generalizations of the DL problem: the  $q$ -Diffie-Hellman Inverse [12] and the Generalized Diffie-Hellman [6]. In both cases, the adversary finally needs to compute a group element. In contrast, in the PDL problem, the adversary is tasked with computing a degree- $t$  polynomial. To the best of our knowledge, this is the first time that the problem PDL (given in Eq. (3)) is explicitly defined and a NIZK PoK is presented for it. However, it is

worth noting that it has been implicitly used in previous VSS schemes [9, 10, 16, 25]. In Feldman VSS scheme [16], given a set of commitments  $\{c_j\}_{j=0}^t$ , one can compute  $g^{f(i)}$  for arbitrary value of  $i$  using the formula  $g^{f(i)} = \prod_{j=0}^t c_j^{i^j}$ . In [10], Cascudo and David also developed a sigma protocol for a variant of  $R_{PDL}$ . In this variation, they utilize different generators, specifically  $F_i = g_i^{f(i)}$ , instead of  $F_i = g^{f(i)}$ . However, when examining the proof of special soundness in their sigma protocol [10, Proposition 1], certain steps are unclear. Notably, there is an absence of a definitive statement and reduction to a hardness assumption. In other words, their proof of special soundness lacks an extraction algorithm. Furthermore, in their work [10], they introduce a probabilistic check protocol for  $R_{PDL}$  and specify that they have no prover. In a general sense, their check protocol uses locally computable checks based on [9]. In this approach, verifiers employ a random codeword from the dual code of the Reed-Solomon code, which was used in the statement. However, in essence, their check protocol can be seen as a non-interactive proof scheme that, in comparison to our NIZK *proof of knowledge* scheme  $\pi_{PDL}$ , achieves *soundness*. Tab. 3 compares the performance metrics for our proposed NIZK proof of knowledge  $\pi_{PDL}$  and compares it with the probabilistic check protocol from [10].

**Table 3.** A comparison of NIZK PoK  $\pi_{PDL}$  with Cascudo and David’s probabilistic check protocol for  $R_{PDL}$  [10].  $n$ : # Elements in the statement,  $t$ : degree of the witness polynomial,  $E_{\mathbb{G}}$ : Exponentiation in group  $\mathbb{G}$ ,  $M_{\mathbb{G}}$ : Multiplication in  $\mathbb{G}$ ,  $\mathcal{PE}$ : degree- $t$  Polynomial Evaluation,  $\mathcal{H}$ : Hashing,  $|\mathbb{Z}_q/\mathbb{G}|$ :  $\mathbb{Z}_q/\mathbb{G}$  element size,  $|\pi|$ : proof size,  $|stat|$ : Statement size.

| Proof Schemes        | Prover                                              | $ \pi  +  stat $                    | Verifier                                               |
|----------------------|-----------------------------------------------------|-------------------------------------|--------------------------------------------------------|
| Check Protocol [10]  | $n E_{\mathbb{G}} + n \mathcal{PE}$                 | $n  \mathbb{G} $                    | $n E_{\mathbb{G}} + n \mathcal{PE} + n M_{\mathbb{G}}$ |
| $\pi_{PDL}$ , Fig. 6 | $n E_{\mathbb{G}} + n \mathcal{PE} + 1 \mathcal{H}$ | $t  \mathbb{Z}_q  + n  \mathbb{G} $ | $2n E_{\mathbb{G}} + n \mathcal{PE} + 1 \mathcal{H}$   |

## 6 $\Pi_S$ : A Novel PVSS Scheme from DL

Building upon the new VSS scheme  $\Pi_F$  (from Sec. 4.1) and leveraging the new NIZK PoK scheme  $\pi_{PDL}$  (from Sec. 5), we present a novel PVSS scheme in this section. It offers a more efficient alternative to Schoenmakers’ scheme [25]. For further context, we have provided an overview of Schoenmakers’ construction [25] in App. B, while here we introduce the new scheme.

**$\Pi_S$ : An Efficient Alternative to Schoenmakers Scheme.** Let  $g$  be a random generator of group  $\mathbb{G}$ . In the initialization step, a party  $P_i$  generates a secret key  $s_i \leftarrow \mathbb{Z}_q$  and registers  $h_i = g^{s_i}$ , as its public key.

In a PVSS scheme, the dealer encrypts the shares under the public keys of the parties and subsequently proves the validity of these encrypted shares. This means ensuring that all the encrypted shares are distinct evaluations of a unique degree- $t$  polynomial  $f(X)$ . Next, we show that building upon  $\Pi_F$  and incorporating the NIZK PoK  $\pi_{PDL}$  (shown in Fig. 6) for  $n \geq 2t + 1$ , we can develop a more

efficient PVSS scheme, designated as  $\Pi_S$ . In  $\Pi_S$ , instead of committing to the shares, the dealer initially encrypts the shares  $f_i$  under the public key  $h_i$  by computing  $y_i = h_i^{f_i}$  for  $i = 1, \dots, n$ . Subsequently, the dealer employs a minimally modified version of the prover from  $\pi_{PDL}$ . This modified prover operates with the inputs  $(h_1, \dots, h_n, 1, \dots, n, y_1, \dots, y_n)$  instead of  $(g, 1, \dots, n, y_1, \dots, y_n)$ , and generates a NIZK proof  $(d, z(X) = r(X) + df(X))$ . In this adapted version of  $\pi_{PDL}$ , the prover, for  $i = 1, \dots, n$ , sets  $c_i = h_i^{r(i)}$  instead of the original protocol's  $c_i = g^{r(i)}$ . At the end, the dealer discloses  $\{y_i\}_{i=1}^n$  as the encryptions of shares and  $\pi_{share} := (d, z(X))$  as the proof<sup>8</sup>. It's important to note that, in  $\Pi_S$ , similar to Schoenmakers' scheme [25], the secret is equal to  $g^{f(0)}$ .

*Verify.* Given  $(\{h_i, y_i\}_{i=1}^n, d, z(X))$ , to verify the shares, a *public* verifier first checks if  $z(X)$  is a degree- $t$  polynomial. If so, it checks if  $d = \mathcal{H}(y_1, \dots, y_n, \frac{h_1^{z(1)}}{y_1^d}, \dots, \frac{h_n^{z(n)}}{y_n^d})$ , and outputs either **true** or **false**.

*Reconstruct.* The reconstruction phase can be done in the same way as in Schoenmakers' PVSS scheme [25], which we summarized before. Note that, as in Schoenmakers' scheme, to reconstruct the secret  $g^{f(0)}$ , the parties do not learn and use the values of  $f(i)$ , rather than  $F_i = g^{f(i)}$ . Also, they do not expose their secret keys, and party  $P_i$  can reuse his/her key pair  $(h_i, s_i)$  in several runs of the PVSS scheme. The description of  $\Pi_S$  is summarized in Fig. 7.

*Security.* The security of  $\Pi_S$  can be proven in the random oracle model through some modifications in the proof of Theorem 5.1 and by referencing [25, Theorem 1, 2] under the PDL and Decisional Diffie-Hellman (DDH) assumptions.

**Theorem 6.1 (Security of PVSS Scheme  $\Pi_S$ ).** *Under the PDL and Decisional Diffie-Hellman (DDH) assumptions, the VSS scheme  $\Pi_S$  (outlined in Fig. 7), is a secure PVSS scheme against a static adversary in the random oracle model. That is, (i) the Reconstruct protocol results in the secret distributed by the dealer for any qualified set of shareholders, (ii) any non-qualified set of shareholders is unable to recover any (partial) information on the secret.*

*Proof.* We need to first show that for any group of  $t + 1$  honest parties (referred to as a qualified set), the reconstruction protocol outlined in Fig. 7 results in a unique secret  $g^{f(0)}$ , distributed by the dealer. To begin, akin to the proof of Theorem 5.1, for proving the special soundness of the interactive variant of the NIZK proof scheme employed during the sharing phase, we can argue as follows. Given two acceptable transcripts of the (interactive) protocol, denoted

<sup>8</sup> It's important to note that this variant of  $\pi_{PDL}$  shares similarities with the sigma protocol proposed in [10], but with two key differences. In our case, we make an assumption that  $n \geq 2t + 1$  (as opposed to their protocol where it's  $n > t$ ) and crucially we require at least  $t + 1$  of the public key owners to be honest and not collude with the dealer. Under these assumptions, we are able to prove the special soundness of this variant, and given the secret keys of  $t + 1$  honest parties, construct an efficient extraction algorithm that extracts a PDL witness from the prover (i.e., the dealer). For more details, please refer to the proof of Theorem 6.1.

**Initial:** As in **Π**, given the generator  $g$  for  $\mathbb{G}$ , parties register a PK  $h_i := g^{s_i}$ .

**Share:** Given  $\{h_i\}_{i=1}^n$ , the parameters  $n$  and  $t$ , to share  $f_0$ , the dealer acts as follows:

- Samples a uniformly random polynomial  $f(X)$  of degree  $t$  with coefficients in  $\mathbb{Z}_q$ , subject to  $f(0) = f_0$ .
- For  $i = 1, 2, \dots, n$ : sets  $y_i := h_i^{f_i}$  as the encryption of  $f_i = f(i)$ .
- Runs a minimally modified version of the prover of  $\pi_{PDL}$  (outlined in Fig. 6) with  $(h_i, i, y_i)_{i=1}^n$  and obtains  $\pi_{share} := (d, z(X) = r(X) + df(X))$ .
- Broadcasts  $\{y_i\}_{i=1}^n$  as the encryption of shares, and  $\pi_{share}$  as the proof.

**Verify:** Given  $\{h_i\}_{i=1}^n$ , ciphertexts  $\{y_i\}_{i=1}^n$ , and the proof  $\pi_{share} := (d, z(X))$ , a verifier first checks if  $z(X)$  is a degree- $t$  polynomial. If so, it checks if  $d = \mathcal{H}(y_1, \dots, y_n, \frac{h_1^{z(1)}}{y_1^d}, \dots, \frac{h_n^{z(n)}}{y_n^d})$  and returns **true** or **false**.

**Reconstruct:** Based on the reconstruction approach outlined in Fig. 2 and [25] parties proceed as follows.

1. They first use their secret key  $s_i$ , and obtain their share  $F_i := g^{f_i}$  from  $y_i$  by computing  $F_i = y_i^{1/s_i}$ . Then, they publish  $F_i$  plus a NIZK proof that the value  $F_i$  is a correct decryption of  $y_i$ . To this end, the party  $P_i$  needs to prove knowledge of an  $s_i$  such that,  $(h_i = g^{s_i}) \wedge (y_i = F_i^{s_i})$ , which can be done by the NIZK proof scheme for the DLEQ relation (given in Fig. 1).
2. Then, given any  $t + 1$  valid values of  $F_i$ , w.l.o.g. for  $i = 1, \dots, t + 1$ , the secret  $g^{f(0)}$ , can be obtained by Lagrange interpolation,

$$\prod_{i=1}^{t+1} F_i^{\lambda_i} = \prod_{i=1}^{t+1} (g^{f_i})^{\lambda_i} = g^{\sum_{i=1}^{t+1} f_i \lambda_i} = g^{f(0)} = g^{f_0},$$

where  $\lambda_i = \prod_{j \neq i} \frac{j}{j-i}$  is a Lagrange coefficients.

**Fig. 7. Π<sub>S</sub>:** An efficient PVSS scheme from discrete logarithm.

as  $(c_i, d, z(X))$  and  $(c_i, d', z'(X))$  for  $i = 1, \dots, n$ , from the verification equation, we know that

$$h_i^{z(i)} = c_i(y_i)^d, \quad h_i^{z'(i)} = c_i(y_i)^{d'} \quad \text{for } i = 1, \dots, n.$$

This implies that,

$$h_i^{z(i)-z'(i)} = y_i^{d-d'} \Rightarrow y_i = h_i^{\frac{z(i)-z'(i)}{d-d'}} \quad \text{for } i = 1, \dots, n.$$

Then, if all  $n \geq 2t + 1$  of the checks in the verification process successfully, given the reconstruction protocol detailed in Fig. 7, any set of  $t + 1$  honest parties can decrypt  $\{y_i\}_{i \in Q, |Q|=t+1}$ , as  $F_i := y_i^{1/s_i}$ , and rewrite the last equation as below,

$$g^{z(i)-z'(i)} = F_i^{d-d'} \Rightarrow F_i = g^{\frac{z(i)-z'(i)}{d-d'}} \quad \text{for } i \in Q, |Q| = t + 1.$$

Now, since  $z(X)$  is a degree- $t$  polynomial, and since from  $f_i := \frac{z(i)-z'(i)}{d-d'}$  for  $i \in Q, |Q| = t + 1$ , we obtain  $t + 1$  *distinct* evaluations of a degree- $t$  polynomial  $\frac{z(X)-z'(X)}{d-d'}$ , therefore an extractor can use  $\{f_i\}_{i \in Q, |Q|=t+1}$  and reconstruct (extract) a *unique* degree- $t$  polynomial  $f(X)$ , which is a witness for the  $R_{PDL}$  relation (or PDL problem). This implies that, any set of  $t + 1$  honest parties,

can use their individual (decrypted) shares  $F_i := y_i^{1/s_i}$ , employ Lagrange interpolation (as in Fig. 7), and evaluate a unique degree- $t$  polynomial  $f(X)$  in the *exponent* for  $i = 0, 1, \dots, n$ . By evaluating  $g^{f(X)}$  at point 0, they can obtain a unique secret value  $g^{f(0)}$ .

Regarding unpredictability, it's important to first note that directly breaking the encryption used in the PVSS scheme implies breaking the Computational Diffie-Hellman (CDH) assumption. Because, given  $g, h_i = g^{s_i}, y_i = h_i^{f_i} = g^{s_i f_i}$ , an adversary would need to compute  $g^{f_i}$ . It is not a difficult task to show that if an adversary  $\mathcal{A}$ , manages to compute  $g^{f_i}$  with some success probability, we can construct another adversary  $\mathcal{B}$  which employs  $\mathcal{A}$  as a subroutine and breaks the CDH assumption with the same success probability. However, this alone does not show that parties cannot obtain partial information about the secret  $g^{f_0}$ . Furthermore, we show that the view of up to  $t$  parties is simulatable. To achieve this goal, a simulator proceeds as follows. W.l.o.g., it first samples  $f(1), \dots, f(t)$  randomly from  $\mathbb{Z}_q$  and sets  $F_1 = g^{f(1)}, \dots, F_t = g^{f(t)}$ , and  $y_1 = h_1^{f(1)}, \dots, y_t = h_t^{f(t)}$ , where  $h_1, \dots, h_t$  are public keys of the  $t$  parties. Then, he samples  $F_{t+1} = g^{f_{t+1}}$  randomly, without knowing  $f_{t+1}$ . Since the point  $f_{t+1} = f(t+1)$  is only given implicitly, we cannot compute the point  $f(t+2), \dots, f(n)$ . It suffices, however, that we can compute  $F_{t+2} = g^{f_{t+2}}, \dots, F_n = g^{f_n}$  by Lagrange interpolation, which also yields the remaining shares. The simulator, now deviates from the protocol by computing the public keys  $h_i$  of parties  $\{P_i\}_{i=t+1}^n$  as  $h_i := g^{w_i}$  for random  $w_i \in \mathbb{Z}_q$ . Then, the simulator sets  $y_i = F_i^{w_i}$  for  $i = t+1, \dots, n$ . This leads to obtain  $h_1, \dots, h_n$  and  $y_1 = h_1^{f(1)}, \dots, y_n = h_n^{f(n)}$ , as required. Next, we note that the underlying proof scheme (i.e., a variant of  $\pi_{PDL}$  from Fig. 6) is honest-verifier zero-knowledge in the interactive case (and ZK in the non-interactive case). Akin to the proof of Theorem 5.1, given the (simulated) statement  $\{h_i, y_i\}_{i=1}^n$  and the challenge value  $d$ , the simulator can sample a random degree- $t$  polynomial  $z'(X)$  and set  $c'_i := h_i^{z'(i)} / y_i^d$  for  $i = 1, \dots, n$ . This results in a simulated transcript which under Decisional Diffie-Hellman (DDH) assumption is indistinguishable from the real view of up to  $t$  parties.

Note that the statement that parties cannot get any partial information from  $(h_i^{s_i} = g^{s_i f_i}, h_i = g^{s_i})$  about the random secret  $s_i$  and  $f_i$  holds under the assumption that ElGamal encryption is semantically secure, which is known to be equivalent to the DDL assumption. Recall that in ElGamal cryptosystem, given the public key  $(g, h = g^f)$ , an encryption of message  $m = 1$  is equal to  $(h^s, g^s)$ , where  $s$  is a random value from  $\mathbb{Z}_q$ .  $\square$

*Efficiency.* Compared to Schoenmakers' scheme [25] and its variants introduced in [9],  $\Pi_{\mathbf{S}}$  offers a better efficiency in general. However, it's worth noting that by applying the same optimization as used in  $\Pi_{\mathbf{S}}$  to reduce the proof length, the unpacked version of Cascudo and David's scheme [10] can achieve a performance level on par with  $\Pi_{\mathbf{S}}$ . For a detailed comparison, please refer to Table 1.

Recently, the developers of Dock Network have also presented an efficient Rust implementation of the  $\Pi_S$  PVSS scheme<sup>9</sup>.

## 7 Conclusion

We introduced  $\Pi$ , as a unified framework for building computational VSS schemes based on Shamir secret sharing [26] that works in the honest-majority setting, achieves optimal resilience, and does not necessarily require a homomorphic commitment, rather than a secure commitment scheme and a random oracle.

Leveraging  $\Pi$ , we proposed three VSS schemes, so-called  $\Pi_F, \Pi_P, \Pi_{LA}$ , and a PVSS scheme, labeled  $\Pi_S$ , which each satisfies different properties.  $\Pi_F$  and  $\Pi_P$  are two RO-based alternatives to the well-known VSS schemes proposed by Feldman [16] and Pedersen [23], while offering a faster verification and reconstruction.  $\Pi_{LA}$  is another instantiation of  $\Pi$  in the quantum random oracle model which compared to the recent VSS scheme proposed by Atapoor, Bagheri, Cozzo, and Pedersen [1] it is slightly more efficient in terms of both computational and communication costs.  $\Pi_S$  is a variation of Schoenmakers' construction [25] and represents a highly efficient PVSS scheme.

We evaluated the empirical performance of our proposed VSS schemes  $\Pi_P$  and  $\Pi_{LA}$  via a prototype implementation, and compared them with Pedersen [23] and ABCP [1] constructions. Our asymptomatic comparisons and implementation results confirm that in general the proposed constructions, outperform the state-of-the-art VSS schemes in the honest-majority setting. Particularly,  $\Pi_{LA}$  can be an attractive scheme for post-quantum threshold cryptography.

We instantiated  $\Pi$  with DL-based and hash-based commitments. However, it is general enough to be instantiated with different commitment schemes, particularly those that are based on PQ-secure cryptographic assumptions.

As a tool for  $\Pi$ , we proposed a novel NIZK proof scheme that might be independently interesting. Specifically, we have defined an extended version of the discrete logarithm relation over *polynomials*, named  $R_{PDL}$ , and presented a new variant of Schnorr's NIZK proof of knowledge scheme for the  $R_{PDL}$  relation. We think that this new NIZK Proof of Knowledge scheme for the  $R_{PDL}$  relation can be a useful tool for the development of more efficient threshold protocols based on Shamir secret sharing. As an example, we have already incorporated it within PVSS scheme  $\Pi_S$ .

At the end, we highlight that the notable efficiency and general nature of the new framework makes it a valuable tool for constructing more efficient VSS schemes and revisiting a wide range of threshold protocols (e.g. DKG protocols, threshold signatures, threshold decryption, and more). Delving into the details of such protocols lies beyond the main scope of this paper. Future research can explore the further applications of  $\Pi$  and integration of new VSS schemes and the new NIZK proof scheme into various cryptographic protocols.

<sup>9</sup> See [https://github.com/docknetwork/crypto/tree/main/secret\\_sharing\\_and\\_dkg/src/bagheri\\_pvss](https://github.com/docknetwork/crypto/tree/main/secret_sharing_and_dkg/src/bagheri_pvss).



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## A Security Proof of $\Pi_F$

Next, we specialize Theorem 3.1 for the case of  $\Pi_F$  and summarize its proof.

**Theorem A.1 (Security of  $\Pi_F$  VSS Scheme).** *Let  $(g_1, g_2)$  be two random group generators and  $\mathcal{H}$  be a random oracle. Then, under DL assumption, the protocol given in Fig. 3 is a VSS scheme in the honest-majority setting that satisfies verifiability and unpredictability. That is, (i) the Reconstruct protocol results in a unique secret distributed by the dealer for any qualified set of shareholders, (ii) any non-qualified set of computationally bounded shareholders is unable to recover the secret.*

*Proof. Verifiability.* The proof of verifiability follows a similar approach to the general construction presented in Theorem 3.1. Specifically, in this case, after rewinding the dealer and relying on the computational binding property of the Pedersen commitment scheme, we can conclude that  $z(i) - d \cdot f_i = z'(i) - d' \cdot f_i$

with overwhelming probability. As a result,  $f_i = \frac{z(i) - z'(i)}{d - d'}$  for  $i = 1, \dots, n$ . Since  $d - d'$  is invertible modulo  $q$ , given any set of  $t + 1$  valid shares, an extractor can extract a *unique* degree- $t$  polynomial from the dealer. Consequently, at the end of the **Reconstruct** phase, any set of  $t + 1$  honest parties can reconstruct a *unique* degree- $t$  polynomial and output a unique value  $f_0$ .

*Unpredictability.* To prove unpredictability, we show that if, given  $\pi_{share}$  and the shares of  $t$  corrupted parties, i.e.,  $\{f_i\}_{i \in Q}$  where  $|Q| \leq t$ , an adversary  $\mathcal{A}$  can recover the shared secret  $f_0$ , then we can build a new adversary  $\mathcal{B}$  that can use  $\mathcal{A}$  as a subroutine and break the Discrete Logarithm (DL) assumption. We provide the reduction for the case  $(n, t) = (3, 1)$ , but it can be easily generalized to the general case.

The adversary  $\mathcal{B}$  acts as follows:

- Let  $y_1 = g_1^{f_1}$  and  $g_1$  be the DL challenge values provided to adversary  $\mathcal{B}$ . In order to obtain  $f_1$ , adversary  $\mathcal{B}$  proceeds as follows:
- It first samples a random value  $\alpha$  and sets  $g_2 = g_1^\alpha$ . It also samples two random values  $f_2, f_3$  from  $\mathbb{Z}_Q$  and sets  $y_2 = g_1^{f_2}$  and  $y_3 = g_1^{f_3}$ .
- Then, it samples a random degree-1 polynomial  $z(X)$  and a random challenge value  $d$ , and sets

$$c_1 = y_1 \left( \frac{g_1^{z(1)}}{y_1^d} \right)^\alpha, \quad c_2 = y_2 \left( \frac{g_1^{z(2)}}{y_2^d} \right)^\alpha, \quad c_3 = y_3 \left( \frac{g_1^{z(3)}}{y_3^d} \right)^\alpha.$$

- Next, it programs the random oracle to  $d = \mathcal{H}(c_1, c_2, c_3)$ .
- Then, it sends  $(g_1, g_2, c_1, c_2, c_3, d, z(X))$  and  $f_2$  (or  $f_3$  in case the third party is corrupted) to adversary  $\mathcal{A}$ .
- Adversary  $\mathcal{A}$  then returns the secret  $f_0$ .
- Finally, adversary  $\mathcal{B}$  computes  $f_1 = \frac{f_0 + f_2}{2}$  as the response to the challenge DL instance  $(g_1, y_1 = g_1^{f_1})$  and returns it.

We observe that the success probability of  $\mathcal{B}$  is equal to the success probability of  $\mathcal{A}$ . This shows that the VSS scheme achieves verifiability and unpredictability under the DL assumption.  $\square$

## B Overview of Schoenmakers PVSS Scheme

In Crypto 99, Schoenmakers [25] proposed a PVSS scheme, based on Feldman's scheme, which allows a dealer to encrypt the shares under the public key of the parties, and then generate a publicly-verifiable non-interactive ZK proof to show that the secret sharing and encryptions are done correctly.

Let  $g, h$  be two random generators of the group  $\mathbb{G}$ . In the initialization step, a party  $P_i$  generates a secret key  $s_i \leftarrow \mathbb{Z}_q$  and registers  $y_i = g^{s_i}$ , as its public key. Then, given  $n$  and  $t$ , to share a *high-entropy* secret  $f_0$ , the dealer of Schoenmakers' construction proceeds as follows:

1. Sample a uniformly random degree- $t$  polynomial  $f(X) := f_0 + a_1X + \dots + a_tX^t$  with coefficients in  $\mathbb{Z}_q$ , subject to  $f(0) = f_0$ .
2. For  $i = 1, 2, \dots, n$ : set  $f_i := f(i)$  and  $y'_i = y_i^{f(i)}$ .
3. Set  $c_0 = h^{f_0}$  and  $c_j = h^{a_j}$  for  $j = 1, 2, \dots, t$ .
4. Let  $x_i = \prod_{j=0}^t c_j^{i^j}$ , for  $i = 1, 2, \dots, n$ . Then, the dealer shows that the encrypted shares  $y'_i$  are consistent by producing a proof of knowledge of the unique  $f(X)$ ,  $1 \leq i \leq n$ , satisfying:  $x_i = h^{f(i)} \wedge y'_i = y_i^{f(i)}$ .
5. To generate the proof for above relation, the dealer uses an extended version of Chaum-Pedersen PoK scheme for DLEQ [13] and acts as follows:
  - (a) For  $i = 1, 2, \dots, n$ , it samples  $r_i \leftarrow \mathbb{Z}_q$ , and sets  $a_i = h^{r_i}$  and  $b_i = y_i^{r_i}$ .
  - (b) Using Fiat-Shamir transform, feeds  $\{a_i, b_i, x_i, y'_i\}_{i=1}^n$  into the random oracle  $\mathcal{H}$ , and obtains a challenge value  $d \in \mathbb{Z}_q$ .
  - (c) For  $i = 1, 2, \dots, n$ : computes  $z_i = r_i - d \cdot f_i \pmod q$ .
6. Publish  $\pi_{share} := (h, c_j, y_i, y'_i, d, z_i)$  for  $0 \leq j \leq t$ , and  $1 \leq i \leq n$ .

*Verify.* To verify the shares, given  $\pi_{share} := (h, c_j, y_i, y'_i, d, z_i)$  for  $0 \leq j \leq t$ , and  $1 \leq i \leq n$ , the verifier acts as follows:

- For  $1 \leq i \leq n$ : computes  $x_i = \prod_{j=0}^t c_j^{i^j}$ .
- For  $1 \leq i \leq n$ : using  $(h, d, x_i, y_i, y'_i, z_i)$ , computes  $a_i$  and  $b_i$ , as follows

$$a_i := h^{z_i} x_i^d, \quad b_i := y_i^{z_i} (y'_i)^d$$

and checks if the hash of  $\{a_i, b_i, x_i, y'_i\}_{i=1}^n$  matches the challenge value  $d$ . If so returns **true**, otherwise returns **false**.

*Reconstruct.* To reconstruct the secret  $g^{f_0}$ , the parties proceed as follows.

1. They first use their secret key  $s_i$ , and obtain their share  $F_i := g^{f_i}$  from  $y_i$  by computing  $F_i = y_i^{1/s_i}$ . Then, they publish  $F_i$  plus a NIZK proof that the value  $F_i$  is a correct decryption of  $y_i$ . To this end, the party  $P_i$  needs to prove knowledge of an  $s_i$  such that,  $(h_i = g^{s_i}) \wedge (y_i = F_i^{s_i})$ , which is done using Chaum-Pedersen [13] proof system for DLEQ (described in Fig. 1).
2. Then, given any  $t+1$  valid values of  $F_i$ , w.l.o.g. for  $i = 1, \dots, t+1$ , the secret  $g^{f(0)}$ , can be obtained by Lagrange interpolation,

$$\prod_{i=1}^{t+1} F_i^{\lambda_i} = \prod_{i=1}^{t+1} (g^{f_i})^{\lambda_i} = g^{\sum_{i=1}^{t+1} f_i \lambda_i} = g^{f(0)} = g^{f_0},$$

where  $\lambda_i = \prod_{j \neq i} \frac{j}{j-i}$  is a Lagrange coefficient.